Design and Control of a Solar-Powered Automated Greenhouse Using IoT Monitoring

Mechatronic Project 478
Final Report

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Executive summary

Title of Project
Design and Control of a Solar-Powered Automated Greenhouse Using IoT Monitoring
Objectives
Design and build a small-scale greenhouse capable of measuring and controlling all the necessary elements to sustain amicable living conditions for various plants while enabling the user with wireless monitoring capabilities. The greenhouse must be powered by solar energy.
What is current practice and what are its limitations?
Current domestically applicable greenhouses rely heavily on active monitoring and care, this requires significant labour and time and can still not achieve ideal living conditions for some plants.
What is new in this project?
This project introduces a system capable of wirelessly monitoring multiple parameters within the greenhouse, including temperature, humidity, soil moisture, and air composition and enables effective control of these parameters within user-defined ranges. The integration of solar power helps to offset the system's energy requirements, reinforcing its sustainability.
If the project is successful, how will it make a difference?
It will introduce the ease of automation and wireless monitoring to domestic greenhouses, demystifying the struggles of home farming while encouraging self-sufficiency. By promoting food independence, it can reduce reliance on the national food supply chain, empowering individuals and communities to grow their own fresh produce sustainably.
What are the risks to the project being a success? Why is it expected to be successful?
The system may not be efficient enough at controlling the desired parameters such as temperature and humidity. There may be delays due to time management issues. However, through thorough research, modular design and allowing buffer periods for unexpected delays, the project shouldn't be able to fail.
What contributions have/will other students made/make?
No other students have contribute to this design.
Which aspects of the project will carry on after completion and why?
Further research on the economic feasibility of the project may be conducted.

What arrangements have been/will be made to expedite continuation?
Thorough documentation of the design process will allow for continuation or improvements to be made. Modular design should allow for ease of maintenance as well as allow for the upgrading of any components if necessary.

Acknowledgements

I would like to begin by sincerely thanking my supervisor for proposing a project that I feel deeply passionate about, and for providing the guidance and support that allowed it to grow into what it has become. I also wish to acknowledge all the staff in the M&M Engineering Department at Stellenbosch University who assisted in many aspects of the project. In no paticular order: Mr Cobus and Ferdi Zietsman, Mr Llewellyn Kordom, Mr Nathi HLwempu, Mr Kevin Neaves, Charl Adonis, Mrs. Paulse Simorne and Mr Cobus Samuels. This project could not have taken the shape it did without their dedication, expertise, and passion for engineering and their respective crafts.

ECSA Outcomes Self-Assessment

Table 2: ECSA Graduate Attributes Mapping

ECSA Outcome	Addressed in Sections:
GA 1. Problem solving: Demonstrate competence to identify, assess, formulate and solve convergent and divergent engineering problems creatively and innovatively.	• Section 3 the problem definition defines and assesses the main engineering problems addressed by the project. • Section 3.5 outlines the generation and comparison of possible approaches to solution. • Appendix A details the evaluation and selection of optimal concepts through decision matrices.
GA 2. Application of scientific and engineering knowledge: Demonstrate competence to apply knowledge of mathematics, basic science and engineering sciences from first principles to solve engineering problems.	• Appendix G includes mathematical analysis and modelling of key design elements, incorporating heat loss, structural analysis, wind loading, fan sizing and power calculations based on first-principle engineering relations and real world application.
GA 3. Engineering Design: Demonstrate competence to perform creative, procedural and non-procedural design and synthesis of components, systems, engineering works, products or processes.	• Section 3 documents the overall design process and conceptual development, guided by design methodology adapted from (Dieter and Schmidt). • Appendix A presents decision matrices for the evaluation of alternative subsystem concepts. • Section 4 provides the final system structure, integration of subsystems, and detailed design documentation. Appendix F.1 includes dimensioned CAD drawings of the frame assembly. • Section 6 outlines test procedures and results for both subsystem and system-level performance evaluation.

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Table 2 – continued from previous page

ECSA Outcome	Addressed in Sections:
GA 5. Engineering methods, skills and tools, including Information Technology: Demonstrate competence to use appropriate engineering methods, skills and tools, including those based on information technology.	• Section 3 documents system and component design using structured design methodologies adapted from (Dieter and Schmidt). • Section 4 includes CAD modelling performed in Autodesk Inventor. • Figure 5.1 shows digital wiring schematics created using electronic design automation (EDA) software. The ESP32 microcontroller was programmed using the PlatformIO IDE with version control maintained through GitHub. • Construction tools included soldering, drilling, laser cutting and 3D printing, alongside use of relays, an H-bridge motor driver, solenoid valves, and various sensors integrated with IoT platforms. Power was supplied through a solar array, charge controller, battery, and buck converter. • Sections 6.2 and 6.6 detail data analysis and visualization conducted using MATLAB and Python.
GA 6. Professional and technical communication: Demonstrate competence to communicate effectively, both orally and in writing, with engineering audiences and the community at large.	• Project Proposal • Progress Report and Presentation • Safety Report • First Draft and Final Report • Final Presentation
GA 8. Individual, Team and Multidisciplinary Working: Demonstrate competence to work effectively as an individual, in teams and in multi-disciplinary environments.	• Although primarily requiring individual effort and responsibility, collaboration across disciplines was essential for component sourcing and fabrication, as described in Section 4.8.4. • Appendix E discusses collaboration within the multidisciplinary environment of the M&M rooftop laboratory, including coordination with safety officers and laboratory managers for risk assessment and compliance.

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ECSA Outcome	Addressed in Sections:
GA 9. Independent Learning Ability: Demonstrate competence to engage in independent learning through well-developed learning skills.	• Section 2 covers literature review and market research across domains such as controlled agriculture, solar energy integration, and automation, extending beyond the standard Mechanical and Mechatronic Engineering curriculum at Stellenbosch University. • Section 4.8 demonstrates self-directed learning through the design and integration of an off-grid solar power system. • Section 6.6 applies independent Python programming for data visualization and analysis. • Appendix G includes heat loss calculations based on (Çengel and Boles, 2019) and buoyancy effect estimations using the stack effect formulation from (American Society of Heating, Refrigerating and Air-Conditioning Engineers, 2021), showing applied learning beyond formal coursework.

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Variables

I	Moment of Inertia	[kgm^2]
E	Youngs Modulus	[Pa]
v	Poisons Ratio	[ ]
P	Dynamic Pressure	[Pa]
D	Flexural Rigidity	[Nm]
p	Density	[kg/m^3]

Vectors and Tensors

$\vec{v}$	Velocity

Subscripts

h	Height
t	Thickness
l	Length
b	Breadth

Abreviations

PAR	Photosynthetic Active Radiation
PPFD	Photosynthetic Photon Flux Density
DLI	Daily Light Integral
CO_2	Carbon Dioxide
ppm	parts per million
RH	Relative Humidity

VWC	Volumetric Water Content
DC	Direct Current
AC	Alternating Current
PV	Photo Voltaic
MPPT	Maximum Power Point Tracking
ADC	Analog to Digital Conversion
GPIO	General Purpose Input Output
GND	Ground
VCC	Voltage Supply
JST	Japanese Solderless Terminal
IoT	Internet of Things
IDE	Integrated Development Environment
TCP	Transmission Control Protocol
IP	Internet Protocol
API	Application Programming Interface
HTTP	Hypertext Transfer Protocol
TX/RX	Transmit/Receive
HVAC	Heating Ventilation Air Conditioning
HAF	Horizontal Airflow
CFM	Cubic Feet per Minute
DoD	Depth of Discharge

Chapter 1

Introduction

1.1 Background

As food insecurity increases and available farmland decreases due to deforestation and urbanization (Olagunju, 2015), small-scale greenhouses offer a practical way to grow food in limited spaces. Greenhouses help protect crops from harsh weather and allow for better control of growing conditions, which can improve yield, product quality and lengthen market availability (Vox *et al.*, 2010). By using transparent materials to trap heat and light, they create a more stable environment for plants. However, maintaining this environment can be labour intensive and often requires regular monitoring to manage temperature, humidity, and watering.

1.2 Objectives

- Design a small-scale greenhouse that can monitor and adjust key environmental factors such as temperature, soil moisture, and humidity.
- Integrate wireless data logging with a dashboard to enable users to access the data from anywhere in real time.
- Allow users to input and control target conditions based on the specific requirements of different plant types.
- Integrate a solar power system to enable off-grid operation.
- Test and evaluate the performance of the greenhouse system to assess its effectiveness and reliability in maintaining suitable growing conditions.

1.3 Motivation

Greenhouses provide ideal growing conditions but require constant monitoring and adjustments, making them labour-intensive and inaccessible for many. By reducing manual effort and enabling automation as well as customization for different plants, this project aims to empower users to grow food efficiently and sustainably with minimal effort. Additionally, the use of renewable resources such as solar power promotes energy independence and environmental sustainability, aligning with global efforts to combat food insecurity and climate change. This project bridges technology and agriculture, making greenhouse farming accessible, efficient, and eco-friendly.

1.4 Methodology

This project adopts the engineering design methodology outlined by Dieter and Schmidt, providing a structured framework to guide the development of the electronic greenhouse. (Dieter and Schmidt) This methodology encompasses several key phases:

1. **Problem Definition and Need Identification:** This initial phase emphasizes understanding the problem's scope and identifying the specific needs the design aims to fulfill.
2. **Concept Generation:** Developing multiple design concepts that address the identified needs and constraints. Multiple potential solutions are brainstormed without immediate judgment, encouraging a broad exploration of ideas.
3. **Concept Selection:** Here, the various concepts are evaluated against predefined criteria to select the most promising solution.
4. **Detailed Design:** Precise specifications for all components are established, including materials, dimensions, and tolerances, preparing the design for manufacturing.
5. **Design Evaluation:** Testing the designed system to determine its functionality and performance.

By following this systematic approach, the project ensures a thorough exploration of design alternatives and a robust development process for the electronic greenhouse.

1.5 Use of AI Tools

AI tools were used only in the writing of this report to enhance language clarity and grammar, ensuring technical information is conveyed effectively.

Chapter 2

Literature review and market research

2.1 Environmental Requirements for Optimal Plant Growth

Different plant species require a careful balance of light, temperature, humidity, and water supply to achieve optimal growth. The following sections outline the key environmental parameters that must be monitored and controlled to create ideal growing conditions within the greenhouse.

2.1.1 Temperature

Plants regulate transpiration and cooling through specialized organs known as stomata located on the surface of leaves. These pores can open or close to control the release of water vapor, helping the plant manage water loss and internal temperature (Xu *et al.*, 2024). As ambient temperatures rise, open stomata allow for increased evaporation, aiding in the cooling process. However, excessive heat can lead to excessive water loss and stress. While optimal temperature ranges vary depending on the plant species and its growth stage, greenhouse temperatures are generally best maintained between 15 °C and 30 °C to support healthy plant development (Koşan *et al.*, 2020).

2.1.2 Light

Light indirectly influences water loss in plants by triggering the opening of stomata, which enables gas exchange and CO_2 uptake for photosynthesis (Kochetova *et al.*, 2022). However, excess light can cause overheating and increased transpiration, so it must be carefully managed along with temperature. Some plants are sensitive to photoperiodism, meaning their flowering depends on day length. In greenhouses, adjusting light exposure can control flowering times, allowing for year-round yield (Torres *et al.*).

To evaluate light conditions within greenhouses, specific terminology and measurement units are used to describe both the quantity and quality of light that plants receive. The spectrum of light between 400 and 700 nanometres (nm) is recognized as **photosynthetically active radiation (PAR)**, the range most efficiently used by plants for photosynthesis (Sena *et al.*, 2024). Monitoring this specific light range is essential for optimizing plant productivity.

A key metric within the PAR range is the **photosynthetic photon flux density (PPFD)**. PPFD refers to the number of photons, within the PAR spectrum, that strike a square meter of plant surface each second. It is expressed in micromoles per square meter per second ($\mu mol \cdot m^{-2} \cdot s^{-1}$). PPFD measurements help determine if the lighting levels in the greenhouse are sufficient to support healthy plant growth. Insufficient PPFD may lead to slow growth or poor productivity due to inadequate light. Excessively high PPFD can result in photoinhibition or physical damage to plant tissues (Hortitech Help Center, n.d.).

To relate PPFD to the more commonly used unit of *lux*, an approximate conversion can be applied The exact factor depends on the spectrum of the light source, but for white LEDs, a typical approximation is:

$$\text{PPFD } (\mu\text{mol}\cdot\text{m}^{-2}\cdot\text{s}^{-1}) \approx \frac{\text{Illuminance (lux)}}{54} \tag{2.1}$$

where 1 $\mu mol \cdot m^{-2} \cdot s^{-1}$ corresponds to roughly 54 lux. This relationship allows lux sensor readings to be interpreted in terms of usable light for photosynthesis. (Ahn *et al.*, 2017) Ideal PPFD ranges for Seedlings range from 100–300, while vegatative range from 300–600 $\mu mol \cdot m^{-2} \cdot s^{-1}$ (Academy, 2024). Monitoring and balancing light intensity through artificial lighting systems and proper greenhouse design is therefore essential to maintain optimal photosynthetic activity without causing plant stress.

2.1.3 Humidity

Humidity plays a critical role in plant health by influencing the rate of transpiration, the process through which plants absorb water and nutrients from the soil and release water vapor through stomata. In high humidity environments, the air becomes too saturated with moisture, which suppresses transpiration. This reduction in water and nutrient uptake can hinder plant growth and development.

Conversely, in low humidity conditions, transpiration rates can become excessively high due to the dry air. In response, plants often close their stomata to conserve water, but this also limits gas exchange and light absorption, leading to water stress and reduced growth (Nederhoff, n.d.).

To support optimal development, research indicates that the ideal relative humidity (RH) range for most greenhouse crops lies between 30% and 90%. Maintaining this balance helps prevent both under- and over-transpiration, ensuring steady nutrient uptake and healthy physiological function (Drygair Greenhouse Dehumidifiers, n.d.).

2.1.4 Carbon Dioxide

For most greenhouse crops, net photosynthesis increases significantly as CO_2 concentrations rise from ambient levels (approximately 340 ppm) up to around 1,000 ppm, potentially boosting photosynthesis by up to 50%. However, the benefit of CO_2 supplementation varies by crop and may not be economically justifiable under low light conditions or for certain species like tulips and Easter lilies, which show little response (UC Agriculture and Natural Resources, n.d.).

CO_2 enters plants through stomata via diffusion, and its uptake is influenced by external CO_2 concentration as well as factors such as light, temperature, humidity, and water availability. While plants generally grow well at ambient CO_2 levels, concentrations can drop to as low as 200 ppm in sealed greenhouses during the day due to plant uptake. This drop can significantly reduce photosynthesis, sometimes with an effect comparable to the enhancement achieved by raising CO_2 levels to 1,300 ppm.

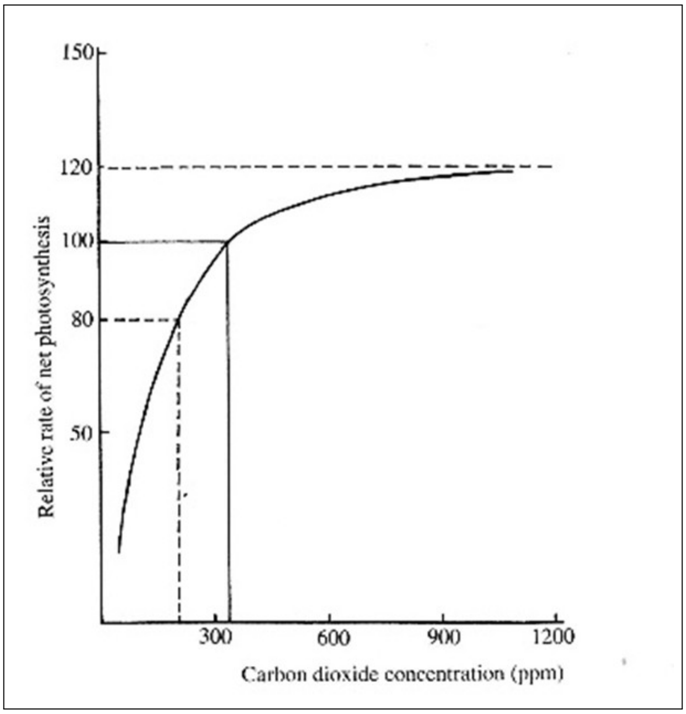


Figure 2.1: Effect of CO_2 concentration on photosynthesis rate. (UC Agriculture and Natural Resources, n.d.).

As seen in the figure 2.4, at low CO_2 concentrations (below approximately 300 ppm), the rate of photosynthesis increases sharply as CO_2 concentration rises. However, beyond approximately 400–500 ppm (close to ambient fresh air levels), further increases in CO_2 concentration have little to no additional effect on the photosynthesis rate, as the curve begins to plateau. Therefore, monitoring and maintaining ambient CO_2 concentrations through simple ventilation can be sufficient to support healthy plant growth, without the need for costly CO_2 enrichment.

2.1.5 Water

A continuous supply of high-quality water is essential for plant transpiration, a process that not only cools the leaves but also facilitates the transport of nutrients from the roots to the leaves and fruits (Olabisi *et al.*, 2022).

To monitor soil moisture effectively, **Volumetric Water Content (VWC)** is commonly used. VWC is expressed as a percentage and represents the volume of water present in a given volume of soil.

$$\text{VWC (\%)} = \frac{\text{Volume of Water (cm}^3\text{)}}{\text{Volume of Soil (cm}^3\text{)}} \times 100 \tag{2.2}$$

This measurable metric provides valuable information for irrigation management, ensuring that plants receive adequate moisture for optimal growth without the risk of overwatering or water stress.

2.2 Structural Design Standards

2.2.1 Achitecture of a greenhouse

Beyond structural integrity, the architectural design of the greenhouse plays a key role in regulating its internal climate. Greenhouses function either to retain heat during cold periods or to exclude heat in hotter seasons, with the aim of maintaining a productive microclimate for plant growth (Ghani *et al.*, 2019). In South Africa's hot summers, a primary design consideration is minimizing heat buildup, making passive and active cooling essential.

One of the most influential architectural elements is the shape of the greenhouse, particularly the roof profile. The geometry directly affects solar irradiance levels inside the structure. In hot, arid environments, the optimal greenhouse form is one that minimizes solar gain during summer while maximizing it in winter, thereby supporting year-round cultivation. Research has shown that the roof angle and shape significantly influence the amount and distribution of solar radiation received (Ghani *et al.*, 2019).

A well-designed greenhouse structure must therefore strike a balance between mechanical resilience (to environmental loads) and thermal efficiency (to manage light and heat), with the shape and orientation of the building serving both functions. There are generally 4 main shapes discussed:

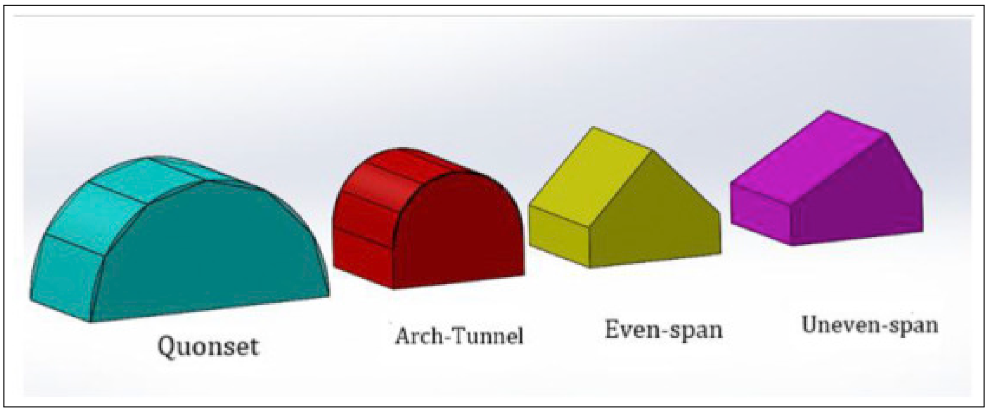


Figure 2.2: Main Greenhouse Shapes. (Sethi, 2009).

Choosing the right shape involves balancing structural load requirements with climate control goals, and often depends on regional weather patterns, crop types, and energy availability.

In terms of geographic suitability, greenhouse performance varies significantly with latitude and shape. For example, at 31°N latitude, even-span and Quonset greenhouses tend to receive more solar radiation in winter and less in summer, which is ideal for regions with significant seasonal temperature variation. However, at higher latitudes such as

50°N, uneven-span greenhouses are more effective for maintaining suitable internal conditions throughout the year. (Sethi, 2009)

The inclination angle significantly affects solar heat gain by altering the amount of radiation absorbed by the greenhouse surface. (Mellalou *et al.*, 2022). As shown in figure 2.3 below, solar energy gaining rates decrease with steeper roof angles such as the 27° inclination angle seen below. While both N–S and E–W orientations perform similarly at 12° inclination, the E–W orientation consistently captures more energy at higher inclinations. This highlights the importance of optimizing both greenhouse orientation and tilt to balance heat gain and internal climate control.

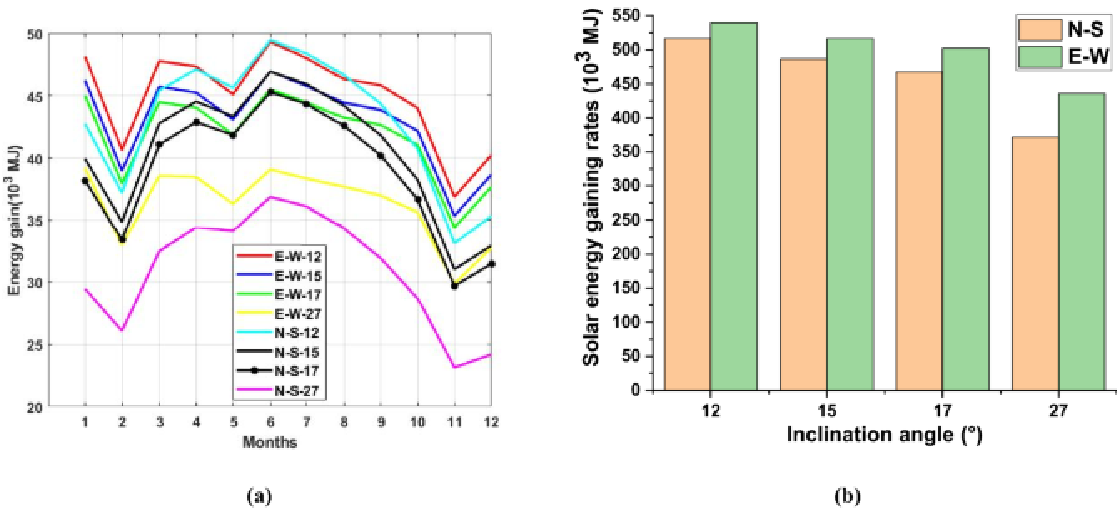


Figure 2.3: Solar radiation gain vs inclination angles (Mellalou *et al.*, 2022).

2.2.2 Sawtooth Design

The sawtooth greenhouse has gained popularity in tropical and subtropical climates due to its unique roof design and natural ventilation system. This roof features several curved, sawtooth-shaped sections, with vertical windows at the highest points to enhance natural ventilation. It uses natural resources like airflow and gravity to regulate the temperature and humidity inside the greenhouse, reducing reliance on external energy sources such as electricity (Insogreen, n.d.). Compared to traditional greenhouses, sawtooth greenhouses provide better ventilation, temperature control, and structural stability, making them ideal for hot, humid regions. As the temperature inside the greenhouse rises, warm air naturally ascends and exits through the vertical windows at the top, while cooler air enters from below, creating a natural airflow cycle. The vertical windows are often controlled by a roll-up mechanism that opens for ventilation and closes for insulation when needed.

Figure 2.4: Sawtooth Greenhouses (Insogreen, n.d.).

2.2.3 Construction techniques

Greenhouses are classified by structure (wooden-, pipe-, and truss-framed) and by framing material. Wooden frames suit small, simple builds; pipe/steel frames add strength for larger spans; and truss systems deliver maximum rigidity for big installations. Materials are chosen for cost, durability, and maintenance: wood is cheap and accessible, however aluminium offers the best mix of strength, corrosion resistance, and low upkeep thus often the long-term choice (Dalai, 2020).

2.2.4 Covering materials

The choice of covering material greatly influences the internal environment, light transmission, durability, and cost of a greenhouse. The most used materials include glass, plastic film, and rigid panels.

Glass Greenhouses: Glass offers high light transmission, good air circulation, and lower internal humidity, which helps reduce disease incidence. However, they are more expensive to build and maintain compared to plastic-covered structures. They introduce additional maintenance challenges and potential issues due to glass's fragile nature.

Plastic Film Greenhouses: Covered with flexible materials such as polyethylene, polyvinyl chloride, or polyester, plastic film greenhouses are cost-effective and require less heating. Their main disadvantage is a shorter lifespan, even UV-stabilized films typically last no more than 4 years.

Rigid Panel Greenhouses: Made from polycarbonate, fiberglass-reinforced plastic, acrylic, or PVC rigid panels, these coverings offer better durability (up to 20 years) and more uniform light distribution. However, the panels are prone to collecting dust and algae, which reduces light transmission over time (Verma *et al.*, 2022).

2.3 Enviromental control

Automated control of environmental parameters such as irrigation, heating and cooling, humidity, and airflow reduces manual workload and contributes to improved plant health.

The following sections outline key industry standards and practices for managing these environmental factors effectively.

2.3.1 Irrigation Systems

The right water system is key to the success of any greenhouse system. As greenhouse plants have a limited soil buffer, precision irrigation systems are vital to ensure high-quality crops and yields (Vegtech Netafim, n.d.). Some of the most used types of irrigation include:

Drip Irrigation: This system delivers water directly to the base of each plant through a network of tubes and emitters, ensuring precise watering and minimizing water waste. By keeping foliage dry, it reduces the risk of fungal diseases. Drip irrigation is highly efficient and can be automated for consistent application (Cedar-Built Greenhouses, n.d.).

Misting System: Misting systems distribute fine water droplets from above to hydrate plants and maintain humidity within the greenhouse. It provides broad and uniform coverage suitable for various crops, particularly delicate seedlings and humidity-loving plants. While effective for moisture control, excessive use can wet plant foliage, potentially elevating the risk of disease. Proper management is essential to mitigate these concerns.

Ebb and Flow Systems: Also known as flood and drain systems, ebb and flow setups periodically flood the growing area with nutrient-rich water and then allow it to drain away. This cyclical process ensures that plant roots receive adequate moisture and nutrients while preventing waterlogging. Ebb and flow systems are commonly used in hydroponic greenhouse operations (FloraFlex Media, n.d.). However, these systems require complex hydraulic components and precise nutrient management, increasing system complexity and maintenance requirements.

2.3.2 Heating and Cooling Systems

Efficient greenhouse climate control uses **passive** and **active** methods (Vegtech Netafim, n.d.). Passive approaches rely on natural processes and strategic design, needing no external energy; active systems use mechanical/electrical equipment requiring external energy inputs.

Passive Heating

This approach leverages solar energy to naturally warm the greenhouse. By incorporating design elements such as proper orientation, insulation, and thermal mass (e.g., water barrels or rock walls), heat is absorbed during the day and released at night, maintaining a stable internal temperature (Kumari *et al.*, 2007).

Active Heating

To compensate for temperature drops, especially during winter nights, various heating systems are utilized. Options include unit heaters such as electric fan heaters, gas heating systems, heat lamps, and solar heating systems. The choice of system depends on factors such as greenhouse size, crop requirements, and energy efficiency considerations.

Passive Cooling

Natural ventilation is employed to facilitate air exchange between the interior and exterior, helping to dissipate excess heat. This can be achieved through operable vents in the roof and sidewalls, allowing warm air to escape and cooler air to enter (Faridi *et al.*).

Active Cooling

In situations where passive cooling is insufficient, active methods like evaporative cooling are implemented. This technique involves using fans and wet pads to cool the air through water evaporation, effectively reducing internal temperatures even during hot periods (Verma, 2022). Integrating both passive and active systems allows for a comprehensive approach to climate control in greenhouses, ensuring optimal growing conditions across varying external temperatures and weather conditions.

2.3.3 Humidity Control

A cost-effective way to achieve humidity control in a controlled environment involves a misting system that sprays mist over the entire crop, thereby increasing humidity (Bluelab, n.d.). It is essential to recognize that relative humidity is strongly influenced by temperature: warm air can hold more moisture than cooler air. As temperatures drop, the relative humidity increases, even if the actual amount of water vapor in the air remains constant. (Lawrence, 2005)

Traditional humidity control methods often involve a combination of heating and ventilation. When humidity exceeds the desired level, greenhouse vents and windows can be opened to expel warm, moist air and allow cooler, drier air to enter. While this effectively reduces humidity, it can also result in unwanted heat loss. As a result, additional heating may be required to maintain optimal growing temperatures. Achieving a balance between ventilation and heating is therefore essential for stable humidity and temperature regulation within the greenhouse environment (Drygair Greenhouse Dehumidifiers, n.d.).

2.3.4 Airflow

Airflow in Greenhouse Environments

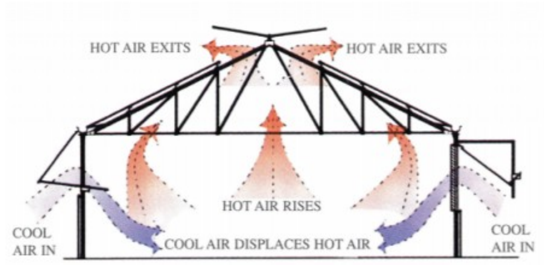
Effective airflow is crucial in greenhouse environments to maintain optimal growing conditions and prevent plant diseases. Proper air movement helps disperse microclimates, preventing localized areas of trapped moisture, and ensures uniform temperature and humidity levels throughout the greenhouse. This uniformity is vital for consistent plant growth and reducing the risk of fungal infections (Drygair Greenhouse Dehumidifiers, n.d.).

Airflow Components in Greenhouses

- **Roof Vents:** These allow hot air to escape from the top of the greenhouse, facilitating natural convection. As warm air rises and exits, cooler air is drawn in, aiding in temperature regulation.

- **Side Vents:** Positioned along the sides, these vents promote cross-ventilation by enabling fresh air to flow horizontally through the structure. This movement helps maintain consistent temperature and humidity levels.
- **Intake and Exhaust Fans:** Mechanical systems that actively move air in and out of the greenhouse. Intake fans draw in fresh, usually cooler air, while exhaust fans expel stale, humid air, preventing the buildup of moisture and heat (GrowersNetwork, n.d.).
- **Horizontal Airflow (HAF) Fans:** These fans circulate air horizontally within the greenhouse, ensuring even distribution of temperature and humidity. By moving air consistently, HAF fans help eliminate microclimates and reduce condensation on plant surfaces, thereby lowering the risk of disease (López *et al.*, 2013).

(a) Greenhouse profile



(b) Ventilation airflow

Figure 2.5: Airflow (GrowersNetwork, n.d.)

The figures above show an active ventilation system displaying the use of side and roof vents to promote healthy air circulation and thermal control. Integrating these components effectively creates a balanced airflow system that supports healthy plant development by regulating environmental conditions and minimizing disease-promoting factors.

2.3.5 Automation and Environmental Control

Automation in greenhouses allows for precise control of environmental conditions such as temperature, humidity, light, and CO_2. Using sensors and controllers, these systems enhance efficiency, minimize manual labor, and support consistent crop development. Depending on the greenhouse's complexity, control systems can range from simple microcontrollers to advanced PLCs and IoT-based architectures (Farooq *et al.*, 2022).

2.3.5.1 Sensors

- **Temperature Sensors**: Monitor the internal temperature of the greenhouse.
  Common types: Thermistors, RTDs, DS18B20 (digital)
- **Humidity Sensors**: Measure the ambient humidity levels.
  Common types: DHT22, SHT31
- **Soil Moisture Sensors**: Detect the moisture content in soil.
  Common types: Capacitive soil moisture sensors, resistive sensors
- **Light Sensors**: Measure light intensity reaching plant surfaces.
  Common types: BH1750, TSL2561, TSL2591

- **CO_2 Sensors**: Monitor carbon dioxide concentration.
  Common types: MH-Z19, Senseair S8

Sensor data is fed into a controller, such as a PLC, Arduino, or Raspberry Pi, enabling automated decision-making for heating, cooling, irrigation, and other critical operations (WebbyLab, n.d.).

2.3.5.2 Controllers

Programmable Logic Controllers (PLC): A Programmable Logic Controller (PLC) is a robust, industrial-grade computer used for controlling and monitoring industrial processes. It operates by receiving inputs, processing the data, and triggering outputs based on preprogrammed logic. PLCs are widely used in greenhouse automation due to their flexibility and reliability. Built with rugged designs and solid-state components, PLCs ensure long-term, reliable performance in harsh environments. The main drawback of PLCs is their higher power requirements and higher cost compared to microcontroller-based systems, which may limit their use in smaller or cost-sensitive projects (Dwinugroho *et al.*, 2021).

Microcontrollers: Microcontrollers are small, programmable devices that can control a wide range of greenhouse automation tasks, such as environmental monitoring and data logging. The following microcontrollers are commonly used in greenhouse automation:

- **Raspberry Pi:** A low-cost, powerful computer board frequently used in Internet of Things (IoT) applications. (Raspberry Pi Foundation, n.d.). It offers greater processing power and versatility than simpler microcontrollers like Arduino and ESP8266. Raspberry Pi is ideal for more complex greenhouse automation systems, including advanced data collection and analysis
- **ESP8266/ESP32:** These low-cost Wi-Fi microchips, developed by Espressif Systems, include built-in transmission control protocol (TCP/IP) stacks and microcontroller capabilities. The ESP8266 and ESP32 are widely used in IoT applications for connecting devices to the internet. They are compatible with the Arduino IDE, making them a popular choice for developing cost-effective, connected greenhouse systems (Espressif Systems, n.d.).
- **Arduino:** An open-source platform based on simple microcontroller chips, Arduino is designed for ease of use and flexibility. It is suitable for basic greenhouse monitoring and control tasks. Arduino's simplicity and large community make it an excellent choice for entry-level projects that require straightforward automation (Yahaya *et al.*, 2019).

These controllers play a critical role in facilitating data logging and remote monitoring as well as enabling automated greenhouse systems to efficiently manage environmental factors such as temperature, humidity, and light for improved crop management.

2.3.6 Wireless Connectivity and Actuation

Wireless Connection:

ThingSpeak can be used with both Raspberry Pi and Arduino to wirelessly upload and visualize sensor data. ThingSpeak is an IoT analytics platform that allows real-time data collection, storage, and visualization from connected devices. Once connected to Wi-Fi, the

Arduino or Raspberry Pi can send HTTP (Hypertext Transfer Protocol) requests to ThingSpeak's API (Application Programming Interface) to upload sensor readings. By integrating ThingSpeak with these controllers, one can effectively monitor and analyze sensor data remotely, making it a valuable tool for greenhouse automation and other IoT applications (MathWorks, n.d.).

Actuators and Switches:

In greenhouse automation, relays are essential components that allow low-voltage control signals to switch high-voltage appliances such as irrigation pumps, lights, and ventilation systems on and off. This ensures efficient and safe operation of the system. These relays can be activated using the low voltage signals sent by the controller (Farooq *et al.*, 2022).

2.4 Energy Efficiency & Sustainability

Solar power lets greenhouses run equipment on a clean, renewable source, cutting costs and environmental impact. Paired with batteries, it keeps systems running through outages and seasons, enabling stable, year-round crop production regardless of outside weather.

2.4.1 Key Requirements for Solar System

Incorporating a solar power system into a greenhouse requires understanding its essential components, each playing a pivotal role in energy capture, conversion, storage, and utilization. The key elements include:

1. **Solar Panels (Photovoltaic Modules):** Solar panels are the primary components that capture sunlight and convert it into direct current (DC) electricity through photovoltaic cells (Krisciunas, 2023).

2. **Inverter:** The inverter transforms the DC electricity generated by the solar panels into alternating current (AC) electricity

3. **Charge Controller with Maximum Power Point Tracking (MPPT):** A charge controller manages the flow of electricity from the solar panels to the batteries, preventing overcharging and prolonging battery life. MPPT is an algorithm included in charge controllers used for extracting maximum available power from a PV module under varying conditions (Eltawil and Zhao, 2013).

4. **Batteries:** Batteries store electricity produced by the solar panels, providing energy during periods without sunlight or during power outages, ensuring continuous operation.

5. **Mounting Systems:** Racking and mounting systems are crucial for securely installing solar panels on the greenhouse structure or the ground. This ensures that panels are positioned at an optimal angle to maximize solar exposure. Standard rails and brackets for various sizes exist on the market (Axe Struct, n.d.).

Understanding and integrating these components effectively is vital for the successful implementation of a solar-powered greenhouse, leading to enhanced energy efficiency and sustainability.

Chapter 3

Concept Design

This section outlines the concept development for the electronic greenhouse. It aims to minimise concept vulnerability by considering many concepts, both good and bad, before choosing one concept that is the best fit to fulfil the desired role.

3.1 Problem Definition

Greenhouses require continuous supervision to maintain optimal growing conditions for plants. This manual oversight can be time-consuming and inconsistent, potentially affecting crop quality and yield. The proposed solution is to develop an electronic greenhouse system capable of monitoring and controlling critical environmental parameters such as temperature, humidity, soil moisture, CO_2 levels, and light intensity. The design should support plant growth while minimizing human intervention. This project focuses on designing and developing a small-scale, automated greenhouse system suitable for educational or personal use. The scope includes:

- Selection and integration of sensors for monitoring environmental variables (e.g., temperature, humidity, CO_2, light, soil moisture).
- Design and implementation of a control system using microcontrollers or PLCs.
- Design and construction of a structural frame using lightweight and transparent covering materials.
- Incorporation of solar power as a primary energy source.
- Implementation of programmable irrigation and ventilation systems.
- Development of a simple user interface with data display and user control.
- Testing and documentation of system performance under typical operating conditions.

The project will not cover large-scale commercial greenhouse operations or complex AI-driven automation but will be designed to allow future scalability or modular upgrades.

3.2 Stakeholder Requirements

Table 3.1 expands on the problem definition given above. The items in the table represent the requirements as stated by the stakeholders. The stakeholders in this project are the end-user, the University of Stellenbosch staff and supervisors, and health and safety officers in compliance with the OSHA Act. The end-user benefits from the smooth operation of the final product, while both the end-user and the university benefit from the design being the most cost-effective option, given that university funding is limited.

Table 3.1: Stakeholder Requirements

Number	Stakeholder	Description	Priority
SR1	End-user	Secure frame and covering material creating thermally insulated space for environment control.	Must-have
SR2	End-user	Monitor environment.	Must-have
SR3	End-user	Data logging capabilities for analysis and decision making.	Must-have
SR4	End-user	Actuation allowing for control of ventilation, airflow, temperature, lighting, and humidity.	Must-have
SR5	End-user	Irrigation system allowing for control of soil moisture content and humidity.	Must-have
SR6	End-user	System powered by solar energy.	Must-have
SR7	End-user, University	Cost-effective.	High
SR8	End-user, University	Energy efficient.	High
SR9	End-user	Environmental parameter allowable ranges wirelessly settable by user with simple user interface.	High
SR10	University	Easy to manufacture.	High
SR11	OSHA	Comply with safety standards.	Must-have

3.3 Engineering Requirements

The stakeholder requirements given in the previous section are expressed as functional requirements (FRs) in Table 3.2 and performance requirements (PRs) in Table 3.3 . Where relevant, the number of the stakeholder requirement that the engineering requirement was derived from, is indicated in the right-hand column. FRs give the actions that the system being designed must peform, as seen by the stakeholders. The PRs are measures of how well the system should perform the functions.

Table 3.2: Functional Requirements

FR Number	Description	Related Stakeholder Requirement
1	Allow transmittance of sunlight	SR1
2	Monitor temperature[1]	SR2
3	Monitor soil moisture	SR2
4	Monitor humidity[1]	SR2
5	Monitor CO_2	SR2
6	Monitor light levels	SR2
7	Display environment conditions	SR3, SR9
8	Process environment conditions	SR3
9	Control temperature	SR4
10	Control humidity	SR4, SR5
11	Control ventilation	SR4
12	Control airflow	SR4
13	Control irrigation	SR5
14	Supply light	SR4
15	Withstand weather conditions	SR1
16	Power the system	SR6, SR8

[1] Refers to both internal and external temperature and humidity for comparison.

Table 3.3: Performance Requirements

PR Number	Description	Target	Range	Unit	Related Stakeholder Requirement
1	Resist Wind speeds[1]	18.81	>18.8	kph	SR1
2	Temperature sensing range[2]	Maximize	-10–50	°C	SR2
3	Soil moisture sensing	0–100	0–100	%	SR2
4	Humidity sensing	0–100	0–100	RH%	SR2
5	CO_2 sensing range [3]	0–1000	0–1000	ppm	SR2
6	Light sensing[4]	0–600	0–600	$\mu mol \cdot m^{-2} \cdot s^{-1}$	SR2
7	Latency in displaying environmental conditions	Minimize	<120	s	SR3
8	Control temperature[6]	User selected[5]	15–30	°C	SR4
9	Control soil moisture content[7]	User selected	50–100	%	SR5
10	Control humidity[8]	User selected	40–90	%	SR5
11	Supplemental lighting intensity[4]	200	100–300	$\mu mol \cdot m^{-2} \cdot s^{-1}$	SR4
12	Energy Consumption	Minimize	0–1200	Wh	SR8
13	Battery storage capacity[9]	24	>24	hours	SR6

[1] (Spark, 2024) Maximum average wind speeds peak at 18.8 kph in Stellenbosch during January.
[2] (Spark, 2024) Stellenbosch climate ranges from extremes of around -10 to 50°C
[3] (UC Agriculture and Natural Resources, n.d.) optimal CO_2 ranges for plant growth
[4] (Academy, 2024). Ideal ranges for Seedlings: 100–300, Vegetative: 300–600 $\mu mol \cdot m^{-2} \cdot s^{-1}$
[5] The System should allow desired ranges to be setable by the user, however suitable testing thresholds are used as follows.
[6] (Koşan *et al.*, 2020). Suitable temperature ranges for a range of plants: 15–30°C.
[7] 50% minimum soil moisture prevents dry soil medium
[8] (Wei *et al.*, 2024). Tolerable humidity is between 40–90% for greenhouses.
[9] The system must maintain power throughout the night.

3.4 Functional Decomposition

The functional requirements outlined in the previous section describe only the high-level actions that the system must perform, as perceived by stakeholders. In this section, these functions are decomposed to identify the internal operations that the system must carry out to meet those requirements.

Figure 3.1 illustrates the flow of energy, information, and/or materials into and out of the system's internal functions. These functions are responsible for consuming, generating, or transforming the various flows to support overall system operation. The primary subsystems and their corresponding functions are numbered from 1.0 to 7.0. These serve as the foundation for the physical decomposition, detailed in Appendix B, developed after the concept evaluation

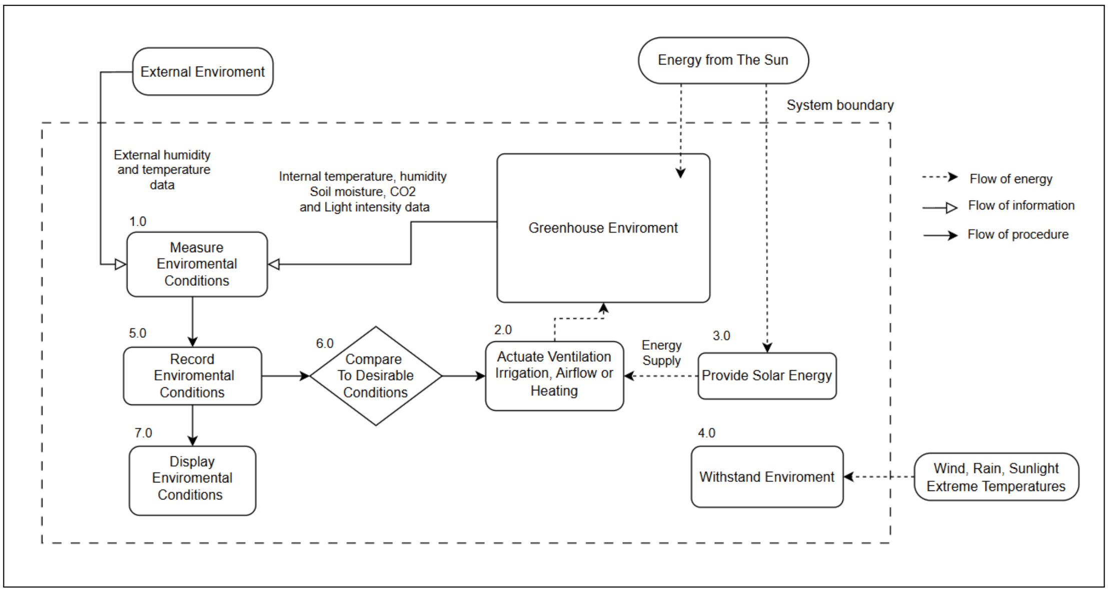


Figure 3.1: Functional decomposition

3.5 Concept Developement

To address the multidisciplinary requirements of the concept design, the system will be partitioned into specialized subsystems. Each subsystem concept is derived from comprehensive market research, focusing on industry-standard operations and adapting these practices to suit the scale and specific needs of a small-scale greenhouse environment. Each subsystem will undergo an independent evaluation to identify solutions that optimally balance functional performance, efficiency, and cost-effectiveness. By analyzing multiple design concepts for every subsystem, this approach aims to mitigate the risk of concept vulnerability and aligns with established systems engineering practices to optimize the overall system architecture. The individual subsystems are defined as follows:

- **Structure of Greenhouse:** The structural concepts are governed by the overall greenhouse shape, which may be an even-span, lean-to, or arch design, as described in Section 2.2.1.
- **Covering Materials:** The covering concepts are defined by material selection, including glass, soft plastic film, or Perspex, as discussed in Section 2.2.4.
- **Active Heating:** The heating subsystem concepts include an electric heater, gas heater, or heat lamp, as outlined in Section 2.3.2.
- **Active Cooling:** The cooling subsystem concepts consist of actuated ventilation, evaporative cooling modules, or misting systems, as referenced in Section 2.3.2.
- **Irrigation:** The irrigation subsystem concepts considered are drip-fed systems, ebb-and-flow systems, or misting systems, as described in Section 2.3.1.
- **Controller:** The controllers considered are microcontrollers namely Raspi , ESP32, or Arduino, as described in Section 2.3.5.2.

3.6 Concept Evaluation

The developed subsystem concepts were evaluated using a weighted scoring method based on key criteria such as efficiency, cost, and manufacturability. Each concept was rated according to these criteria, and the highest-scoring alternatives were selected for detailed design. The full evaluation process and scoring matrices are presented in Appendix A. The final Concept decisions for each subsytem are detailed below.

3.6.1 Structure

For small-scale applications, the lean-to option emerges as the most practical solution, balancing cost, space utilization, and manufacturability, while still offering adequate thermal performance.

3.6.2 Covering Material

Perspex was selected as the covering material due to its durability, ease of fabrication and ability to be cut to shape, and favourable balance between thermal insulation and light transmittance.

3.6.3 Active Heating

An electric heater was selected as it offers efficient, evenly distributed heating, seamless integration with the solar power system, and lower cost and complexity compared to gas heaters or heat lamps.

3.6.4 Active Cooling

Actuated ventilation is the most economical and energy-efficient cooling option, relying only on fans to exchange hot and cool air. However, it can cool only to the ambient external temperature. Combining ventilation with misting enhances evaporative cooling and improves temperature regulation, though it requires a pressurized water supply and humidity control. Evaporative cooling modules were ruled out due to higher cost, power demand, and reduced performance in humid conditions. Thus, a combined ventilation–misting approach offers the best balance of cost, simplicity, and effectiveness.

3.6.5 Irrigation

Misting emerges as the most effective method due to its dual role in both irrigation and environmental control, despite requiring higher water pressure. A misting system with sensor-based control is therefore preferred for optimizing both plant growth and climate regulation.

3.6.6 Controller Selection

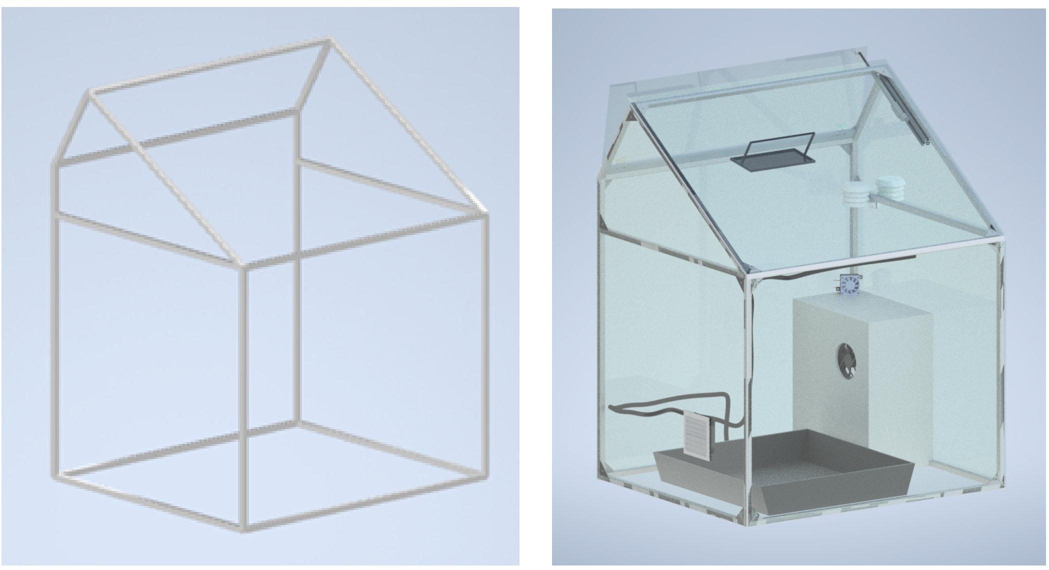
The ESP32 Offers a balance of performance and cost, with built-in Wi-Fi and Bluetooth capabilities, making it ideal for IoT and wireless applications.

Chapter 4

Detailed Design

This section presents the detailed greenhouse design using a structured approach: for each design element, requirements are defined first, followed by technical specifications and implementation details. A key design priority is the roof vent to maximize passive ventilation, which guided the geometry and mechanism choices that follow. Following design reviews, a revised structural design was developed to enable improved operation and manufacturability. The general shape was optimized to accommodate laser cutting processes, with the use of the universities CO2 laser cutter with a maximum available bed size of 1260 mm × 860 mm. The roof geometry was modified to an uneven-span lean-to configuration. While this maintains a steep roof inclination for efficient solar gain, it also allows for better thermal control and ensures that the roof vent can be actuated even when the greenhouse is positioned adjacent to a wall.

Allowing users to place the greenhouse against a thermally massive surface (e.g., a concrete or brick wall) enhances passive heat retention and improves overall energy performance. Additionally, the revised structure introduced more right angles, simplifying manufacturing and increasing the usable internal volume of the greenhouse.



(a) Frame Structure

(b) Final Concept

Figure 4.1: Detailed Design

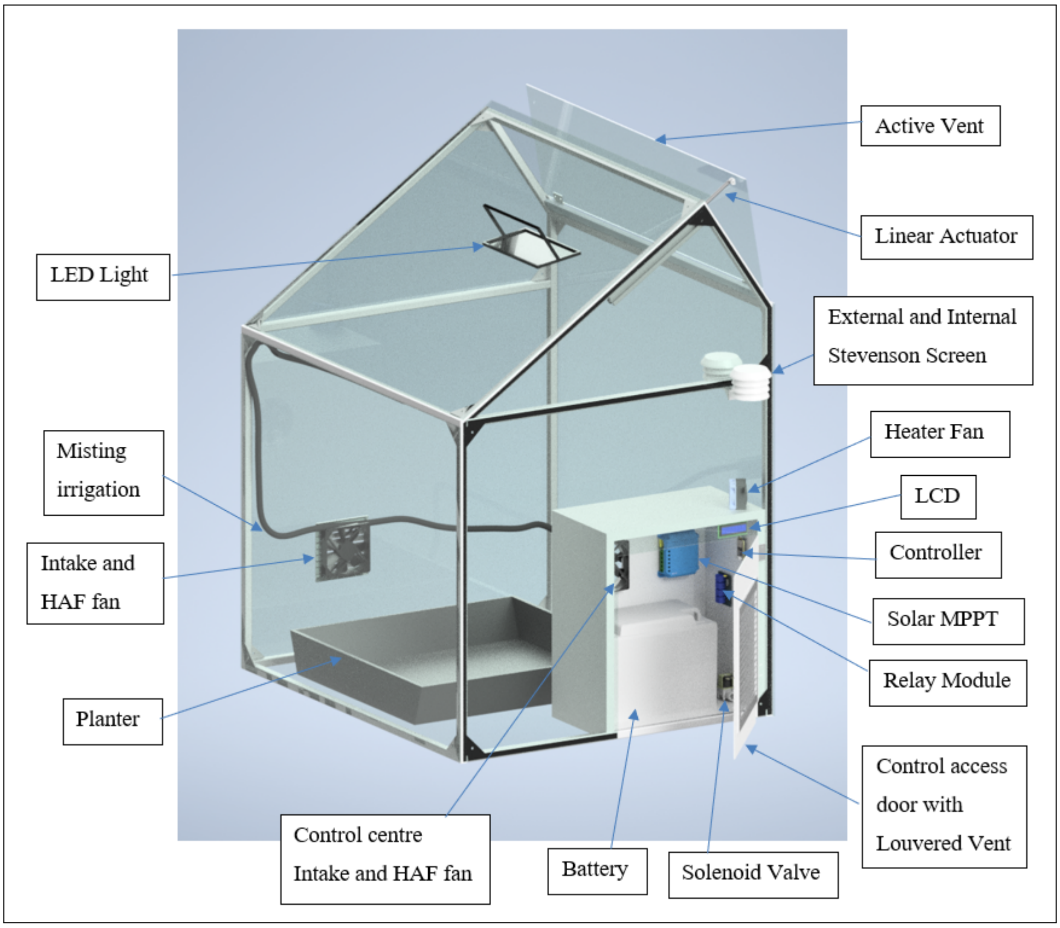


Figure 4.2: Final Concept Definition

The control center protecting all of the electrical components is shifted to the side of the greenhouse and a door is laser cut from the right side Perspex panel allowing easy access for maintenance. This side access preserves the ability to have the greenhouse be leveraged against an existing wall maximizing space efficiency. Hinges on the front Perspex panel enable it to open for convenient access required for planting and harvesting.

4.1 Structure

4.1.1 Frame

The frame must provide a durable structural foundation that supports the greenhouse covering material, withstands environmental loads such as wind and rain, and does so in a cost-effective manner. Aluminum square tubing is selected for its excellent strength-to-weight ratio. Compared to a stainless steel frame, aluminum offers a much lighter structure, making the greenhouse easier to transport or reposition if necessary. Additionally, aluminum is naturally corrosion-resistant, eliminating the need for protective coatings to prevent rust. Standard aluminium extrusions are also widely available and relatively easy to cut and assemble, making it a practical and cost-effective choice for the frame. A structural analysis seen in the calculations in Appendix G confirmed that the frame can withstand its own weight, the weight of Perspex panels, and wind loads simultaneously. The calculated safety factor for the aluminium frame, based on maximum wind loads of up to 40 km/h acting perpendicularly to the side panels, was determined to be approximately 40.1 . The maximum deflection simulated at the apex of the greenhouse amounted to only 0.5 mm, demonstrating excellent structural stiffness and integrity. Detailed engineering drawing of the frame assmbly can be seen in Appendix F.1

4.1.2 Covering material

Perspex was chosen over glass for its combination of machinability, thermal performance, transparency, and non-brittle nature, addressing both construction and long-term durability needs.

4.2 Control Center Design

4.2.1 Electrical Component Housing

A dedicated area is required for the safe and neat storage of all electrical components such as the battery, charge controller, buck convertor, microcontroller, relays and valve. The control center can be made from wood as it is simple to construct and mount parts to, as well as being electrically nonconductive and thermally insulating. Specifically, marine plywood is used as it resists water damage and is made from high-quality hardwood veneers, making it stronger and more resistant to warping compared to standard plywood. Coating it with white exterior-grade paint will further improve water resistance and reflect sunlight, keeping internal components cooler while reflecting more light for the plants. Silicone sealant is applied in and around the control centre to ensure waterproofing and isolate it from the greenhouse environment, protecting components from misting and rain.

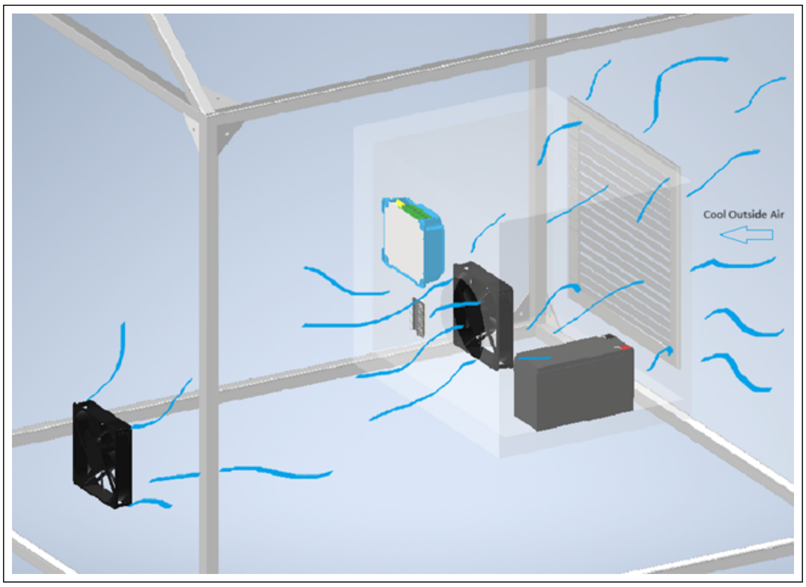


Figure 4.3: Air Intake

4.2.2 Control Center Ventilation

A cutout from the Perspex allows for placement of a louvre vent as seen in figure 4.4. These vents will allow for passive ventilation to keep the control center cool while the shape of the louvre vents will prevent water from entering. This vent is placed on the side of the greenhouse to allow for the placement of the greenhouse against a wall.

For this reason, the control center was moved to the right side of the greenhouse, freeing up more space for plants. Dedicated ventilation prevents overheating, a strategically positioned HAF fan mounted on the enclosure's side creates an airflow path drawing outside air through the large louvred vent, across the electronics, and exhausting it into the greenhouse. This dual-purpose setup cools the control and electrical components while supplying fresh air to the growing environment.

4.2.3 Access Panel

A cutout from the side panel around the control center allows for the placement of a door providing easy access to the electrical components. This door will also incorporate the louvred vent, enabling proper ventilation for the electrical components and acting as the intake for the HAF fan attached to the side of the control center.

4.3 Intake and HAF fans

Intake fans bring in fresh oxygen rich air when needed. To regulate the greenhouse temperature, cool air is drawn in from the base while a top vent opens to allow warm air to escape, promoting natural convection and effective cooling. Industrial greenhouses also rely on HAF fans to promote healthy air movement and prevent stagnation. However, on a smaller scale, dedicated HAF fans are not necessary. For the sake of simple design and using smart controller logic these two actions of air intake and HAF can be combined to be completed by the same two components. In this case two 12V fans on either side of the greenhouse are placed towards the bottom section of the greenhouse. These two fans will allow for air intake when necessary as well as intermediate HAF control at set intervals. The fans are placed slightly offset from each other to promote circular airflow and improve coverage.

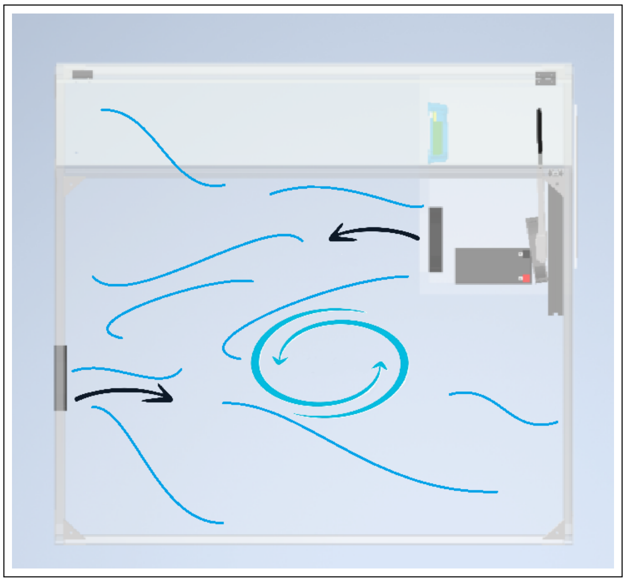


Figure 4.4: Air circulation

Exhaust fans were excluded from the design as the roof vent, having a large surface area, will allow for the escape of a large quantity of air when the air intake and ventilation is active. This exhaust flow is assisted by the buoyancy effect as stipulated in the calculations in Appendix G

4.3.1 HAF Requirements

A common guideline is to size the HAF fans to be able to move 2 CFM for each square foot of greenhouse floor area (Watson *et al.*, 2019). By these standards for the floor area of 0.85 m^2 an HAF capacity of at least 18.3 CFM would be required to ensure healthy air circulation. Two 120 mm PC fans with an average of 70 CFM each will be more than sufficient and will be activated during dedicated HAF periods.

$$A_{\text{floor}} := b \cdot l = 0.85\,\text{m}^2 \tag{4.1}$$

$$A_{\text{floor,ft}} := A_{\text{floor}} \cdot 10.764\,\frac{\text{ft}^2}{\text{m}^2} = 9.1494\,\text{ft}^2 \tag{4.2}$$

$$CFM_{\text{HAF}} := A_{\text{floor,ft}} \cdot 2\,\frac{\text{ft}}{\text{min}} = 18.2988\,\frac{\text{ft}^3}{\text{min}} \tag{4.3}$$

Due to the use of misting systems within the greenhouse care needs to be taken to protect the fans and other components from moisture buildup, for this reason 3D printed covers and louvered vents are designed to protect from droplet buildup on components. Smart placement of misting nozzles and intermediate misting control will also prevent excessive moisture exposure to electrical components.

4.4 Ventilation

4.4.1 Natural Ventilation

An effective ventilation system at the top of the greenhouse will allow for passive cooling using the buoyancy effect of the warm air within the greenhouse. The larger the area of the vent the more effective the air transfer and thus cooling capabilities. A Perspex panel on the roof measuring 1000mm by 270mm is to be actuated to the open position to allow for this exchange. A linear actuator is selected as it is less complicated than installing a motor and shaft and more cost effective. The actuator selected should have a large stroke of at least 100mm to increase the opening area and will need to be rated to lift the weight of the Perspex panel. Calculations seen in Apendix G were completed to ensure that the specification can be met and an example can be seen in the free body diagram below.

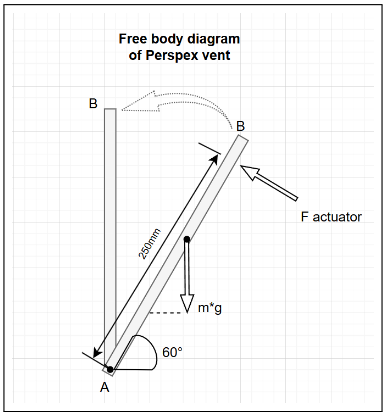


Figure 4.5: Free body diagram of Vent Mechanism

When opened, the vent exposes the greenhouse interior to potential precipitation, which could result in overwatering or water exposure to electrical components. To mitigate this, a rain sensor is mounted on the exterior of the greenhouse and connected to the microcontroller. When moisture is detected, the sensor signals the system to close the vent, thereby protecting both the plants and internal electrical components during periods of rain.

4.4.2 Forced Ventilation

Typical greenhouse design guidelines recommend approximately one complete air exchange per minute (Watson *et al.*, 2019). With a greenhouse volume of 0.77 m^3, this requires two fans with a combined flow rate of at least 0.77 m^3/min. Calculations in Appendix G indicate that the volumetric flow rate of the intake fans must be at least 27 CFM. Accordingly, two PC fans, each averaging 70 CFM, are selected, more than sufficient to exchange the entire greenhouse volumes air within one minute.

4.5 Misting System

Effective greenhouse environmental control requires regulating soil moisture while maintaining relative humidity (RH) within crop-specific setpoints. To avoid the complexity of parallel systems, this design adopts a single misting system coordinated by scheduling and control procedures to handle both irrigation and humidity management. When irrigation or a misting cycle elevates RH, the mechanical ventilation is automatically engaged to exhaust excess moisture and restore RH to the target range. The developed control algorithms for this optimization can be seen in appendix C. A further advantage of the misting system is evaporative cooling, which operates in combination with the greenhouse's passive (roof vent) and active ventilation to limit temperature spikes above desired thresholds.

Misting Nozzles are placed effectively to ensure moisture reaches the roots directly. Up to 4 nozzles are placed slightly above the soil and deliver water straight to the soil medium. Misting nozzles offer adjustable spray patterns, ranging from a targeted drip to an ultra-fine mist. Two additional nozzles will be installed at a higher elevation within the greenhouse to provide broader coverage while still targeting specific zones. Careful positioning will ensure mist is directed away from sensitive electrical components.

4.6 Active Heating System

To achieve effective active heating, a fan heater is selected. Based on the heat loss calculations on the surface area, the greenhouse experiences a convective heat loss of approximately 119 W when there is a 5 °C temperature difference between the internal and external environments.

Given that many plants require a minimum desirable temperature of around 15 °C for optimal growth, the heating system must be capable of compensating for this heat loss. Therefore, a fan heater with a power rating of at least 120 W is required.

A 12,V, 120,W fan heater is provided by the university to meet this requirement, powered directly by the 12,V battery supply. The heater is controlled via a relay module, which is switched by signals from the microcontroller. This setup allows the heater to automatically activate when the internal temperature falls below a user-defined threshold. A hysteresis band is implemented in the code, this prevents fast unnecessary switching, allowing for accurate feedback and extending the lifespan of the heater and relays.

4.7 Environmental Monitoring

The greenhouse uses a network of environmental sensors connected to a central microcontroller to continuously monitor key parameters such as temperature, humidity, soil moisture, light intensity, and CO_2 concentration. These measurements provide real-time feedback on the internal climate conditions, which are essential for optimizing plant health and growth.

The data collected is processed by the microcontroller and used both for local decision-making (e.g., activating ventilation or irrigation systems) and for remote monitoring via IoT platforms. Through cloud-based interfaces, users can access historical data and current conditions, and adjust system settings as needed. The following sections detail the selection and implementation of the microcontroller, the individual sensors, and the data acquisition platforms used in the system.

4.7.1 Controller

The ESP32 microcontroller, developed by Espressif, is selected for its abundant I/O options, making it ideal for managing multiple sensors and actuators through its numerous GPIO pins. Its built-in Wi-Fi and Bluetooth connectivity enables seamless communication with user interfaces, allowing for real-time monitoring of the greenhouse environment.

The ESP32 supports a variety of communication protocols such as I^2C, UART, and SPI, allowing seamless integration with environmental sensors and control modules. Its dual-core processing power ensures efficient real-time data handling and decision-making. Its integration with Arduino's IDE software allows for easy programming capabilities. Additionally, the ESP32's low power consumption and wide community support make it a reliable and cost-effective choice for an embedded greenhouse controller.

4.7.2 Sensors

Multiple sensors enable accurate monitoring of both the internal greenhouse environment and the external ambient conditions. This data informs the control of greenhouse parameters and provides a baseline for comparison between internal and external climates.

The external temperature and humidity sensor will be exposed to direct sunlight and rainfall, while the internal sensors will experience both sunlight and periodic misting. To ensure longevity and reliable performance, protective housings are required. Custom 3D-printed Stevenson screens can shield sensors from water and heat while allowing airflow through their slatted design for accurate readings. White filament is recommended to reflect sunlight and minimize heat absorption. In accordance with standards, the Stevenson screen should be mounted around 1.2m above ground level (Environment and Climate Change Canada, 2017).

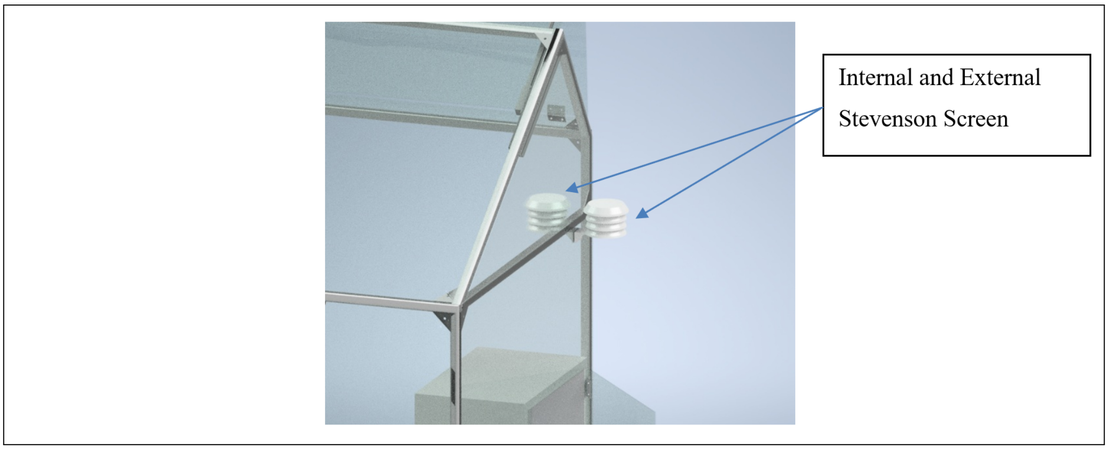


Figure 4.6: Stevenson Screens for sensor protection

The following sections break down each sensor's selection and operation.

4.7.2.1 Temperature and Humidity Sensor

The DHT22 sensor was selected for its balance of accuracy, low cost, and ease of integration. Since it measures both temperature and relative humidity, it reduces the need for separate sensors, streamlining wiring and system design. It communicates with the ESP32 using a single-wire digital interface, making implementation straightforward.

Table 4.1: DHT22 Temperature and Humidity Sensor Specifications

Parameter	**Specification**
Temperature Range	-40 to 80 °C
Temperature Accuracy	±0.5 °C
Humidity Range	0–100 % RH
Humidity Accuracy	±2–5 % RH
Operating Voltage	3.3–5.5 V
Output Signal	Digital (single-wire protocol)

4.7.2.2 Light Sensor

The BH1750 light sensor is chosen for its good balance of accuracy, cost-effectiveness, and ease of integration. It provides precise ambient light measurements in lux, making it suitable for monitoring plant lighting conditions with a simple conversion to PPFD values. It's I^2C communication protocol allows seamless and reliable connection with the ESP32, using only two GPIO pins. The sensor also requires minimal configuration and is supported by well-documented libraries, enabling quick setup and consistent performance in greenhouse environments.

Table 4.2: BH1750 Light Sensor Specifications

Parameter	Specification
Operating Voltage	3.3-5V
Interface	I^2C (7-bit address: 0x23 or 0x5C)
Measurement Range	1–65,535 lx
Resolution	1 lx (High-Resolution Mode)
Accuracy	$\pm$20% at 25°C
Current Consumption	$\approx$0.12 mA (active), 0.01

4.7.2.3 Carbon Dioxide Sensor

The MH-Z19 CO_2 sensor is chosen due to its balance of affordability, accuracy, and ease of integration with microcontrollers like the ESP32. It uses Non-Dispersive Infrared (NDIR) technology to detect CO_2 concentrations, which provides reliable and stable readings with minimal calibration necessary.

The sensor communicates through UART or PWM, offering flexibility in how data is read by the system. Its compact design and low power consumption (typically 33 mA at 5 V) make it ideal for continuous indoor monitoring in environments such as greenhouses. The sensor has a measurement range of 0–5000 ppm with an accuracy of ±(50 ppm + 5% of reading) in the 0–2000 ppm range.

Table 4.3: MH-Z19 CO_2 Sensor Specifications

Parameter	Specification
Communication	UART, PWM
Operating Voltage	3.6–5.5 V
Current Consumption	33 mA (typical)
Measurement Range	0–5000 ppm
Accuracy	±(50 ppm + 5% of reading)
Warm-up Time	3 minutes

4.7.2.4 Soil Moisture Sensor

The soil moisture sensor needs to provide real-time monitoring of substrate hydration levels, enabling the automated misting system to deliver precise irrigation when needed. A generic industrial grade capacitive soil moisture sensor and module is selected for its simplicity, cost-effectiveness, and compatibility with the ESP32 microcontroller.

Table 4.4: SEN0114 Soil Moisture Sensor Specifications

Parameter	**Specification**
Output Signal	Analog voltage
Operating Voltage	3.3–5 V
Interface	1-wire (requires ADC)
Probe Material	Corrosion-resistant alloy
Measurement Depth	Up to 37 mm

The sensor operates by measuring the conductivity between two exposed probes inserted into the soil, where higher conductivity indicates greater moisture content. It provides an analog voltage output (0–3.3 V) corresponding to the soil's moisture level, allowing for real-time monitoring of water content in the growing medium.

The sensors ease of integration and low power requirements (drawing less than 20 mA) make it well-suited for irrigation automation. The sensor only requires one wire for communication and connects to an analog-to-digital converter (ADC) pin on the ESP32. Initial calibration is required to establish the relationship between voltage output and actual soil moisture content. This can be achieved as follows.

Calibration Procedure

1. **Hardware Setup:** Connect the soil moisture sensor's output to an ADC compatible pin (e.g., GPIO 11) set to 12-bit resolution on the ESP32. Ensure the sensor is powered correctly.
2. **Wet Point Calibration:** Place the sensor probes in fully saturated soil (considered 100% moisture). Record the stable ADC reading, R_{wet}.
3. **Dry Point Calibration:** Place the sensor probes in completely dry soil (considered 0% moisture). Record the stable ADC reading, R_{dry}.
4. **Software Implementation:** In the microcontroller code, map subsequent ADC readings (R) to a percentage value using the following linear interpolation formula:

$$\text{Moisture Percentage} = \left(\frac{R - R_{\text{dry}}}{R_{\text{wet}} - R_{\text{dry}}}\right) \times 100\% \tag{4.4}$$

4.7.2.5 Rain Sensor

A rain sensor is mounted on the exterior of the greenhouse to detect rainfall and trigger protective actions. When rainfall is detected, the system is programmed to automatically close the ventilation openings, thereby safeguarding electrical components and preventing overwatering of the plants.

The selected model is an HKD general-purpose rain sensor, which features a large sensing surface to improve droplet detection. The sensor operates by using exposed conductive tracks on the board: when water bridges the tracks, the resistance changes significantly, from approximately 2 MΩ when dry to around 100 kΩ when wet. This variation is processed by the module, which outputs a digital signal (logic HIGH or LOW) to indicate the presence of rain.

4.7.3 Data Acquisition and Monitoring

The ESP32 can collect sensor data and store it locally in .xls format on an SD card. However, to provide a more accessible and user-friendly interface, data will instead be uploaded to a cloud-based platform such as ThingSpeak. ThingSpeak supports monitoring and control of up to 8 fields with a minimum update interval of 15 seconds. The free tier offers up to 3 million messages per year, which equates to approximately 8,000 messages per day. Blynk is another free IoT platform that allows for easy creation of mobile apps to control and monitor hardware remotely. It is fully compatible with the Arduino platform, making it a convenient choice for IoT projects. A Blynk dashboard is thus setup alongside a ThingSpeak dashboard to allow for wireless monitoring and control of the greenhouse environment.

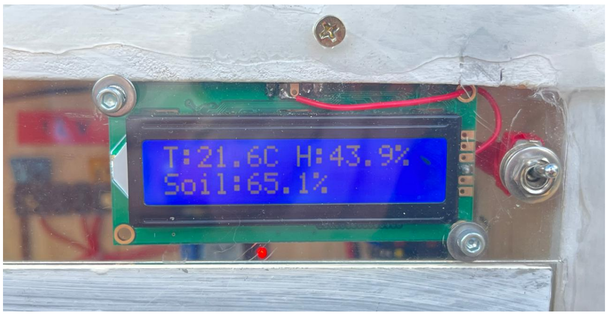


Figure 4.7: LCD screen for local monitoring

Local Monitoring

Local monitoring is useful when a user is on-site near the greenhouse and wants to easily view the current internal conditions. Additionally, in the event of a Wi-Fi outage, the greenhouse can still communicate essential data locally. For this reason, a small LCD screen is used to display key environmental parameters, including temperature, humidity, light levels, soil moisture, and CO_2 concentration. The LCM1602A display is selected for its built-in I^2C communication interface. This allows it to share the same two pins already used by the BH1750 sensor, since I^2C supports multiple devices on a single bus through unique addresses. The LCM1602A operates at address `0x3E`, enabling efficient integration while reducing GPIO pin usage compared to the standard 8-bit parallel interface. The screen cycles through these readings to present the information clearly and efficiently. It is mounted above the control room door and is visible through the Perspex panel, allowing for quick visual access without needing to enter the enclosure. The active screen alongside the all purpose on/off switch can be seen in the figure above.

4.8 Power Supply

The power supply provides energy to the system as a whole and is crucial for the general operation and reliable functioning of the device. Power is supplied by a 12 V battery through the integration of a barrel jack. However, several components require 5 V or 3.3 V for operation. Therefore, voltage conversion is necessary to supply the correct operating voltages. Instead of a traditional linear voltage regulator, a buck converter is chosen for its higher efficiency, especially when stepping down from 12 V. Two fuses are incorporated to safeguard the components against potential failures or voltage spikes. Following the guideline of using 1.5 times the maximum expected current, a 15 A fuse is installed to protect the 12 V line, supplying high-power components such as the solenoid valve and heater. Additionally, a smaller 5 A fuse is placed on the supply line to the buck converter, ensuring protection for the lower-power electronics operating at 5 V and 3.3 V.

4.8.1 Step-down Buck Converter

A buck converter is a type of DC-DC converter that efficiently reduces a higher input voltage to a lower output voltage using high-frequency switching elements (typically a transistor and a diode) along with energy storage components like inductors and capacitors. This method significantly reduces power loss compared to linear regulators, which dissipate excess voltage as heat.

For this application, an MP4560 buck converter module is used. It is a high-efficiency, step-down switching regulator capable of handling input voltages up to 36V and outputting up to 3A. It is configured to provide a steady 5V output, which is distributed via a 5V rail for convenient access by various components. The MP4560 features an integrated high-side switch, built-in thermal protection, and short-circuit protection, making it well-suited for embedded applications.

An LED is included in the 5V power circuit to indicate active power supply, giving the user a visual cue that the system is powered. In accordance with the MP4560's datasheet recommendations, decoupling capacitors are placed on both the input and output to stabilize voltage and suppress noise. These capacitors serve to filter out voltage ripples and transient fluctuations by storing charge and releasing it when needed, ensuring clean and stable voltage to connected components.

4.8.2 Battery Capacity

A battery capacity of at least 89Ah is required to power the average energy usage of the system and its components as shown in the calculations in Appendix G. The calculations provide for an allowable minimum depth of discharge (DoD) of 20% for recommended safe sealed gel battery usage. A battery with a capacity of 100Ah is provided by the university to meet the requirements.

4.8.3 Solar System

A solar system rated to provide at least 240W would be able to charge the battery fully in one day assuming a moderate system efficiency of 80% as shown in the calculations. Thus a single solar panel of at least 240W or two 120W panels can be sourced.

4.8.4 Provided Equipment

The M&M department of engineering has provided a 12V, 100Ah battery which should be sufficient based on the previous DoD calculations. Additionally two solar panels rated at 105W are provided along with a MPPT Skyking Solar Charge Controller. This charge controller will charge the battery from the 2 solar panels which will power all the components.

4.9 Relays

The ESP32 operates with 3.3V digital logic, which is suitable for interfacing with most sensors. However, it cannot directly control components that require higher voltages or draw significant current. To address this, relay modules are used to enable the ESP32 to switch high-power devices using low-voltage control signals. Two 4-channel relay modules have been selected for this project, capable of switching loads up to 250V and 10A, and compatible with 3.3V logic levels. This module will allow the ESP32 to control 12V devices such as fans, the linear actuator, and the heater efficiently and safely.

4.10 Linear actuator and H-Bridge

To actuate the roof vent, a linear actuator is required. The Actuonix L16 S linear actuator is selected for this purpose due to its 12V operating voltage, its availability, and ability to deliver up to 50N of force, verified through the calculations to be more than sufficient to lift the perspex panel. The actuator includes built-in limit switches that stop motion at full extension or retraction. To control both extension and retraction, the polarity of the 12V power supply must be reversed. For this reason, an H-bridge circuit is necessary. Instead of using a dedicated H-bridge driver, the design implements two switching relays wired in an H-bridge configuration. By toggling the relays, the polarity of the 12V supply to the actuator can be reversed, enabling both extension and retraction. Careful coding is essential to ensure that both relays are never activated simultaneously, as this could create a short circuit across the supply. Proper logic sequencing in the microcontroller program guarantees safe operation while maintaining reliable control of the actuator. 3D-printed components can be used to mount the linear actuator to the Perspex panels, providing two degrees of freedom to accommodate slight pivoting during extension and retraction.

4.11 Solenoid valve

A solenoid valve operates using an electromagnetic coil that converts electrical energy into mechanical motion. (Evans, 2019) When the coil is energized, it generates a magnetic field that moves a plunger or valve mechanism, opening or closing the fluid passage. This enables precise electronic control of water flow in irrigation systems.

The solenoid valve is used to automate watering within the greenhouse. It is powered from the 12V supply and controlled via one of the relay channels, allowing the ESP32 microcontroller to activate the valve as needed without directly handling high current loads. The integration of a solenoid valve ensures reliable, repeatable irrigation control, enabling system automation and reducing manual intervention.

Chapter 5

System Integration

This section of the report outlines the plan for integrating both the physical hardware components and the software architecture of the greenhouse system. It begins with the wiring layout of all electronic components, followed by a description of the software framework and control logic that will govern the system operation. This section on system integration demonstrates how the various subsystems, sensors, actuators, and controllers, work together to achieve the overall functionality of the greenhouse. It also provides a clear roadmap for implementation, troubleshooting, and future scalability.

5.1 Wiring

The battery provides a 12V power rail to any components that require it. This 12V supply is connected to the Common (COM) terminals of the relay modules. When activated by the microcontroller, each relay allows current to flow from the COM terminal to the Normally Open (NO) terminal, supplying 12V to the connected device. On the input side of the relay module, the VDD pin must be connected to a 5V power source, and the GND pin to ground. The four input pins labeled IN1–IN4 are connected to general-purpose I/O (GPIO) pins on the microcontroller, specifically GPIO 28, 29, 34, 35, 40, 13 and 8 allowing the relays to be toggled via 3.3V logic signals.

These relays control the 12V-powered components: airflow fans, heater fan, grow light, and solenoid valve. Each of these devices requires a connection to ground on their negative terminal, which is wired to the negative terminal of the battery. A common ground rail ensures all components share the same electrical ground reference. The wiring schematic created using Electronics Design Automation (EDA) software can be seen in the figure 5.1 below.

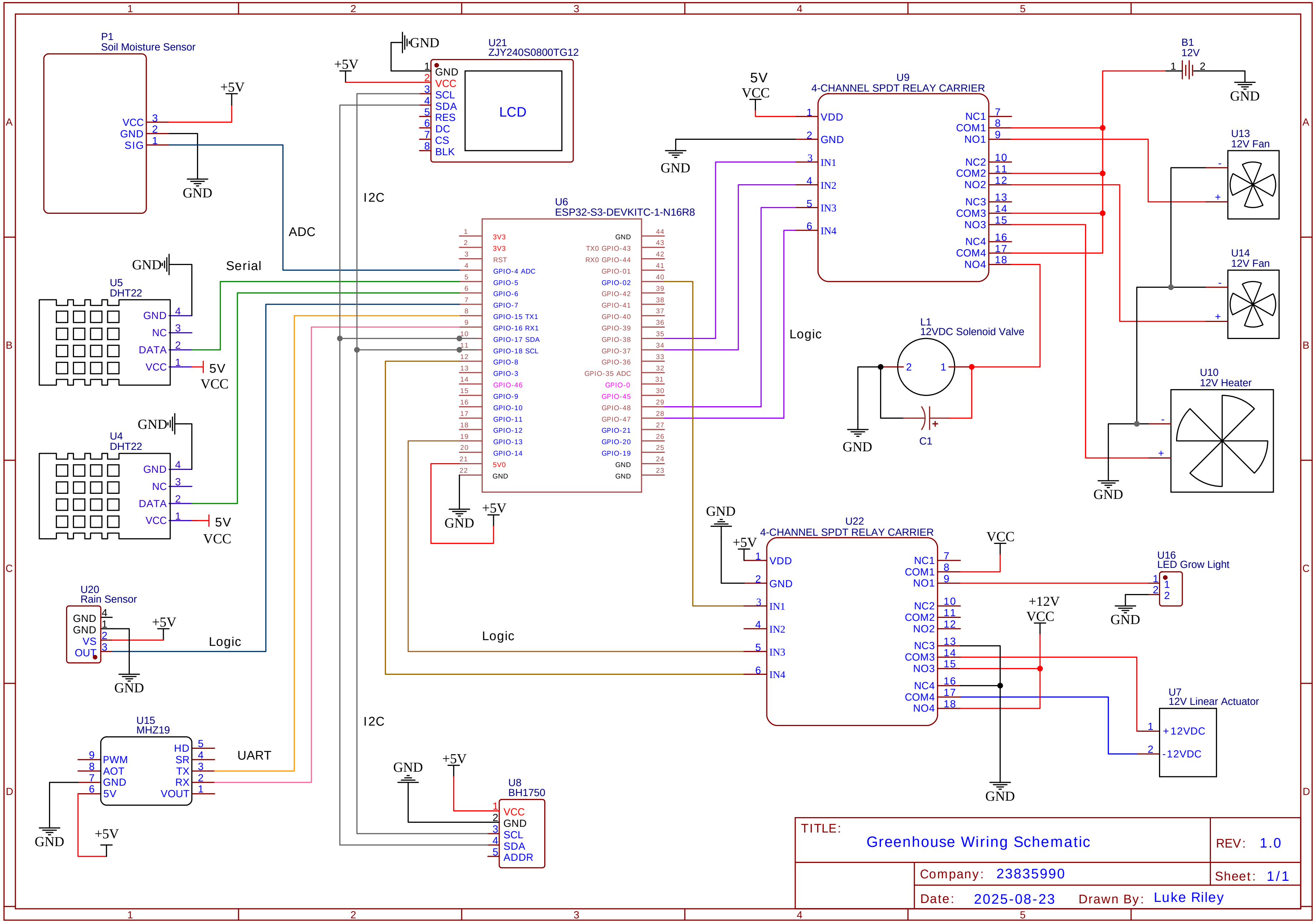


Figure 5.1: Wiring schematic

For the linear actuator, bidirectional motion is achieved by reversing the polarity of the supply voltage using two channels of the relay module. GPIO 8 and GPIO 13 of the micro-controller are connected to input pins IN3 and IN4 of the relay module, respectively. The 12V supply is wired to the Normally Open (NO) terminals of both relays, while the actuator leads are connected to the Common (COM) ports. The Ground connection is wired to the normally closed (NC) contacts of each relay so that, when unpowered, the channels default to ground, protecting the actuator. With this configuration, switching the relays in opposing states alternates the polarity applied to the actuator, enabling both extension and retraction.

Other components and sensors require a 5V supply. A buck regulator is used to step down the 12V input efficiently, avoiding the high energy loss associated with voltage dividers. The regulator's VIN is connected to the 12V rail, and its GND pins are tied to ground. The VOUT pin is connected to a 5V power rail that supplies all 5V-dependent components. Before connecting any loads, the regulator should be calibrated by measuring the output with a multimeter and adjusting the onboard potentiometer until the desired 5V output is reached.

The DHT22 sensors each require a 5V VCC connection, ground to GND, and a data line connected to GPIO 5 and 6 respectively, using digital communication. The soil moisture sensor is similarly powered with 5V to VCC, ground to GND, and its data line connected to GPIO 4 enabling ADC conversion. The BH1750 light sensor communicates via the I^2C protocol and is connected to 5V and ground, with Serial Data (SDA) and Serial Clock (SCL) lines wired to GPIO 17 and 18 respectively. The rain sensor outputs a logical high voltage level of 3,3V when water is detected, accordingly GPIO 7 is set as an input pin to read the logic level. The MHZ19 CO_2 sensor requires a 5V supply to its 5V pin, GND to ground, and Transmit/Receive (TX/RX) lines connected to UART-compatible GPIO pins, GPIO 15 and 16 in this case.

For local display of environmental readings, an I^2C LCD module is used. The module requires 5V on its VCC pin and ground to GND. Communication with the microcontroller uses the I^2C bus: GPIO 17 is configured as SDA and GPIO 18 as SCL. The BH1750 light sensor shares this bus, and additional devices can be added on the same two lines, each addressed by a unique I^2C address. This configuration reduces the number of required connections compared to a parallel LCD, while still enabling reliable display of real-time sensor data.

For low-voltage signal wiring, a crimping tool is used to terminate Japanese Solderless Terminal (JST-XH -2.54 mm) connectors. Their 2.54 mm pin pitch matches standard perfboard/breadboard spacing, simplifying soldering onto prototype boards housing the microcontroller circuitry. These connectors enable quick attachment and detachment of longer wire buses to and from sensors, carrying both power and data.

5.2 Software Framework and Control Logic

The software design aims to provide reliable, automated control of the greenhouse environment by continuously monitoring sensor inputs and actuating components to maintain desired growing conditions. The primary objectives are to ensure stable temperature, humidity, soil moisture, and light levels, while maintaining system efficiency, and ease of monitoring adjustment by the user.

This section outlines the software design framework for the greenhouse system. The accompanying flowcharts illustrate the sequence of tasks and decision-making processes implemented in the programming of the microcontroller. These flowcharts serve to simplify the development of complex control algorithms, particularly those involving multiple variables with competing operational requirements.

In certain situations, control actions are prioritized based on environmental conditions. For example, if the internal temperature exceeds the setpoint while soil moisture levels remain adequate, the misting system may still be activated to assist with cooling requirements. In such cases, temperature regulation is prioritized over soil moisture conservation.

The control logic is structured into distinct modules, arranged in approximate order of priority: Principal Control, Temperature Control, Airflow Control, Irrigation Control, Lighting Control, and Air Quality Control. The principal control flowchart is shown in Figure 5.2, while the subsequent higher level detailed flowcharts for each subsystem are presented in Appendix C.

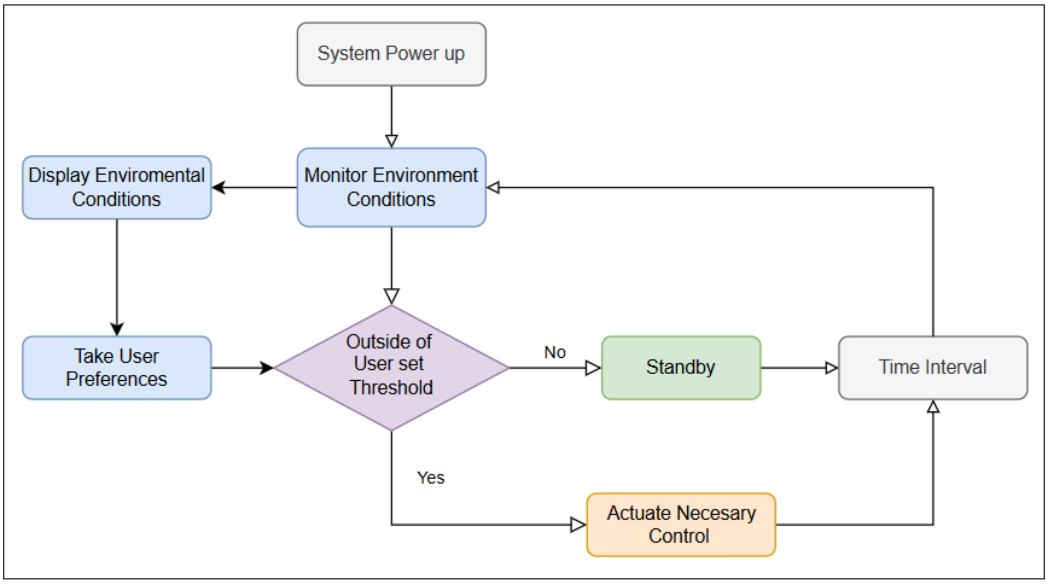


Figure 5.2: Principal Control Flowchart

Principal Control involves monitoring internal and external environmental conditions to determine which components should be actuated in response to undesirable conditions. These environmental readings are wirelessly and locally displayed to the user, who can then set custom threshold values based on the specific requirements of the plants being cultivated. When any measured condition falls outside the user-defined thresholds, the system responds by activating the appropriate control mechanisms. If no user preferences are provided, the system defaults to a general-purpose set of control parameters suitable for common plant types, as presented in table 3.3

Including these flowchart diagrams helps to streamline programming of the microcontroller's control logic by providing a clear implementation framework, reducing the likelihood of debugging issues caused by conflicting decision-making processes. Development was done in PlatformIO IDE alongside the use of GitHub for version control.

Chapter 6

Design Evaluation

This chapter describes the tests used to validate the greenhouse system's setup and operation. It includes performance tests that assess the selection and design of sensors and actuators, solar power supply, the integrated environmental control, and the wireless monitoring and control capabilities. Where applicable, tests are designed and presented following standard experimental methodology. The chapter concludes with a full-system demonstration: growing microgreens over a two-week period.

6.1 Greenhouse Setup

The Full Greenhouse setup can be seen in the figure below. Please see Appendix H for additional pictures.

(a) Greenhouse Setup

(b) PV Panels

Figure 6.1: Greenhouse setup and PV panels.

6.2 Sensor Validation

All sensors were tested to verify their response and accuracy to real environmental changes under controlled conditions. Each was connected to the ESP32 microcontroller with the appropriate pin configuration as detailed in section 5.1, and real-time data was logged at one-second intervals through the serial communication port connected to a PC. The data was then imported into MATLAB for processing and visualization. This setup ensured consistent testing and provided clear insight into each sensor's performance characteristics.

6.2.1 Temperature and Humidity Sensor (DHT22)

Test Objective

The purpose of this test was to verify the accuracy and responsiveness of the DHT22 sensor to real environmental changes in temperature and humidity.

Methodology

1. The sensor was connected to the ESP32 via serial communication, and live readings were recorded.
2. A burst of warm air was directed at the sensor to simulate a rapid change in conditions.
3. Temperature and RH were logged and plotted in MATLAB for analysis.

Results

The plotted data, shown in Figure 6.2, demonstrates the sensor's response to the applied stimulus. A clear rise in temperature was observed immediately after exposure to warm air, accompanied by a corresponding change in humidity.

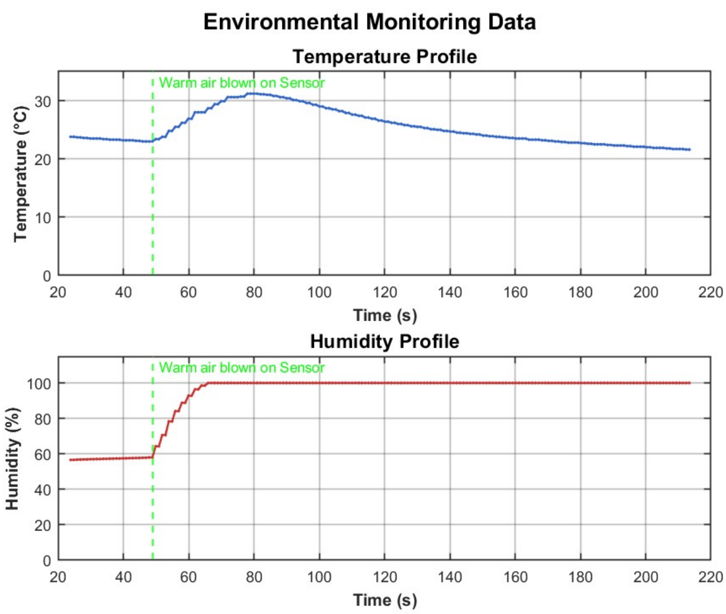


Figure 6.2: DHT22 response to applied warm air (temperature and humidity vs. time).

Discussion

The test confirms that the DHT22 sensor responds promptly to changes in environmental conditions, with smooth and continuous data output. The test showed the expected rise in both temperature and humidity. However, it was noted that the humidity readings temporarily saturated at the maximum percentage due to condensation forming within the sensor. After a short period, the values gradually returned to ambient levels. This behavior is a known characteristic of the sensor under extreme conditions and is not expected to cause issues during normal greenhouse operation, where such rapid condensation is unlikely to occur. Slight response delays were also noted, which are consistent with the DHT22's documented sampling rate and latency.

6.2.2 Summary of Additional Sensor Tests

Following the same testing approach described above, all sensors were validated for responsiveness and reliability. Each test involved controlled exposure to changing environmental conditions, with data recorded via serial communication and analyzed in MATLAB. A summary of key results of each sensor is provided below, as the experimental procedures followed were identical to those of the DHT22 test.

Soil Moisture Sensor

The sensor was connected to the ESP32 microcontroller through ADC and real-time values were logged. The sensor probes were sequentially tested in open air, damp soil, and fully submerged conditions. As shown in Figure 6.3, the readings ranged from approximately 0% in dry air to 80% when submerged, demonstrating clear distinction between soil moisture states. The sensor provided consistent and repeatable readings, confirming its suitability for automated irrigation control. The calibration process will ensure more accurate mapping of percentage values to volumetric water content, but the results demonstrate strong baseline performance for the intended application.

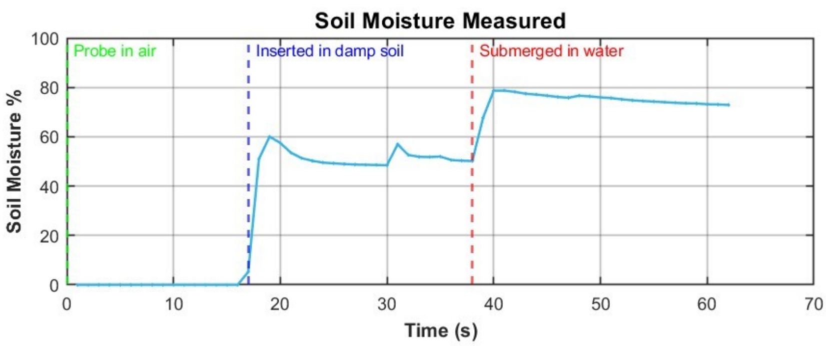


Figure 6.3: Soil moisture sensor response in open air, damp soil, and submerged in water.

Light Sensor

The BH1750 light sensor was connected to the ESP32 via the I^2C interface, and its lux readings were logged in real time. The sensor was exposed to varying illumination using a handheld torch. As seen in Figure 6.4, the sensor responded sharply, rising from

40 lux (ambient) to approximately 200 lux under direct torch illumination, then returning to baseline upon removal of the light source. This confirmed its accuracy and responsiveness to dynamic lighting conditions.

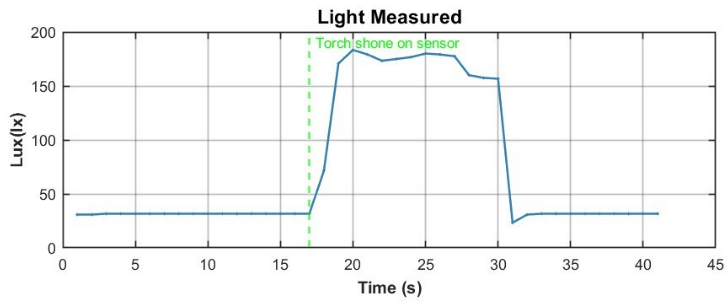


Figure 6.4: BH1750 light sensor response to torch illumination and return to ambient lighting.

CO_2 Sensor

The MHZ19B CO_2 sensor was connected to the ESP32 via UART communication and its readings were logged in real time. The sensors readings were allowed to stabilize at ambient indoor conditions (approximately 1000 ppm) before being exposed to a burst of exhaled air. The concentration rapidly increased to the sensor's limit of 5000 ppm, then decayed back to baseline within two minutes, as shown in Figure 6.5. This validated the sensor's sensitivity, recovery time, and overall suitability for environmental monitoring of air quality.

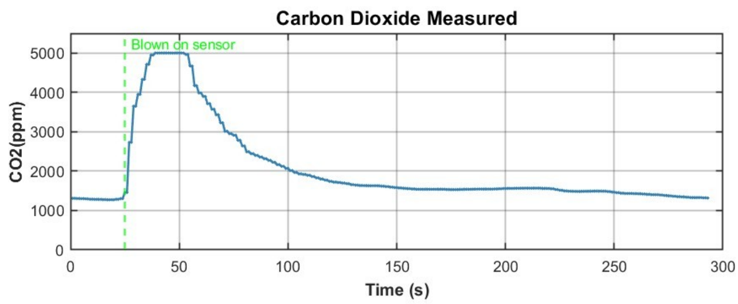


Figure 6.5: CO_2 sensor response to rapid change in air quality.

Summary Discussion

Across all tests, the sensors exhibited expected and reliable responses to environmental variations. Minor latency and saturation effects observed under extreme conditions are consistent with manufacturer specifications and are not expected to affect normal operation. The results confirm that all selected sensors are well-suited for integration into the automated greenhouse control system providing essential monitoring of the various internal and external environmental variables.

6.3 Actuator Validation

Each actuator was connected to its respective relay channel, corresponding to the assigned GPIO pin on the microcontroller. Testing was carried out by first supplying the rated voltage directly to each actuator, followed by validation through relay control. A logical HIGH signal was written to each GPIO pin to trigger the relay, with successful activation confirmed visually via the built-in indicator LED and audibly by the distinct clicking sound of the relay switch. As the relay module operates in active-high mode, actuators were connected to either the Normally Open (NO) or Normally Closed (NC) terminals as required. Once verified, the appropriate 5 V or 12 V supply was connected to the Common (CO) port to power the respective actuators. Table 6.1 summarizes the testing procedures and outcomes for each actuator.

Table 6.1: Actuator Testing and Observations

Actuator	**Test Procedure**	**Observation**
Solenoid Valve	12 V applied directly, then via relay control. Bulk and high-frequency capacitors added to suppress voltage spikes.	Audible switching confirmed; misting nozzles activated and deactivated with relay control.
Linear Actuator	12 V applied with polarity switching, then via relay-controlled H-bridge.	Extension and retraction observed correctly in both directions. Relay control verified reliable actuation.
Supplemental Lighting	12 V applied directly, then switched via relay control.	Purple UV grow light switched on and off reliably under both direct and relay control.
Intake / HAF Fans	12 V applied directly, then via relay control.	Fans powered on successfully; airflow observed. Relay switching confirmed proper operation.

All actuators operated successfully during direct and relay controlled tests. With the mains water supply connected, the solenoid valve successfully controlled the misting system, providing adequate pressure to the misting nozzles. The nozzles were observed to start and stop spraying water in direct response to the relay switching.

During further testing, it was observed that switching the misting solenoid simultaneously with Wi-Fi activity (e.g., ThingSpeak uploads) introduced supply transients that caused the ESP32 to briefly brown-out and reset. The solenoid's inductive kickback and relay contact bounce injected noise onto the 12 V rail, which coupled into the 5 V supply. To mitigate this, a bulk electrolytic capacitor and a high-frequency ceramic capacitor were added at the solenoid driver terminals. These components damped the transient response and provided local energy buffering, resulting in stabler system operation without undesired timeouts or resets.

6.4 Remote monitoring

6.4.1 Objective

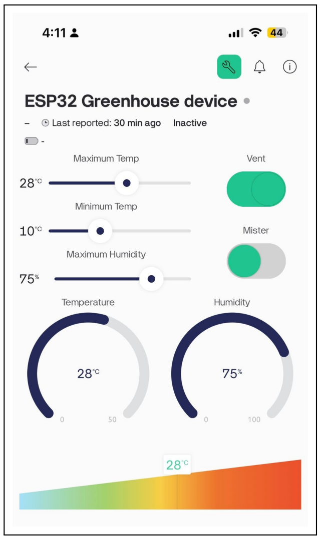
The objective of this experiment was to verify the successful implementation of remote monitoring and user control for the greenhouse system. The aim was to confirm that environmental parameters could be measured, transmitted, stored, and visualized in real time through wireless platforms, and that user-defined setpoints could be remotely updated.

6.4.2 Methodology

The ESP32 was configured to interface with two IoT platforms: *ThingSpeak* and *Blynk*. For remote monitoring, it transmitted internal and external temperature and humidity, soil moisture, CO_2, and light intensity to ThingSpeak at 20-second intervals. Integration used the ThingSpeak API over HTTP and the firmware issued POST requests to `/update` with the channel's write API key and the relevant field values (`field1–field7`) in the payload. A custom ThingSpeak dashboard was then used to visualize the data in real time (Figure 6.6a).

For remote control, the system was integrated with the Blynk IoT platform. A mobile dashboard was created (Figure 6.6b) to enable the user to adjust key control parameters such as temperature and humidity setpoints through sliders and buttons. The ESP32 was connected to Eduroam services and configured with a unique Blynk template and authentication token, allowing remote updates of setpoints for various environmental parameters.

(a) ThingSpeak Wireless Monitoring Dashboard.



(b) Blynk Mobile Control Interface.

Figure 6.6: Remote monitoring and control interfaces

6.4.3 Results

The greenhouse successfully transmitted all sensor data to ThingSpeak without communication interruptions. The online dashboard updated at the programmed 20-second interval, confirming the reliability of wireless data transfer and cloud visualization. Historical data was logged and plotted on the ThingSpeak channel, verifying successful storage for later analysis, as presented in Section 6.6.

The Blynk dashboard enabled real-time modification of environmental setpoints from a mobile device. Commands sent via Blynk were received by the microcontroller with negligible delay. Parameter changes were reflected in the system's behaviour, demonstrating successful two-way communication between the user and the greenhouse.

6.4.4 Discussion

The experiment confirmed that both remote data monitoring and user interaction were successfully achieved. While ThingSpeak proved effective for real-time data visualization and long-term data storage, Blynk offered a more intuitive interface for real-time control adjustments. The combined use of these platforms allows both analytical monitoring and interactive management of greenhouse conditions.

6.5 Solar Power Test

To validate the capability of the solar subsystem to power the greenhouse entirely off-grid, a series of tests were conducted. The procedure was as follows:

1. The battery voltage was measured using a mulimeter with no load connected, showing 12.8 V, indicating a partial state of charge (full charge is approximately 14.8 V).
2. The battery was connected to the Skyking charge controller via the positive and negative terminals. Upon connection, the charge controller powered on and displayed the battery voltage.
3. Using the controller menu, the float voltage was set to 13.7 V, the discharge stop voltage to 10.8 V, and the battery type configured as a sealed gel battery, in accordance with the battery manufacturer's recommended operating conditions.
4. The solar panels were connected using crimped MC4 connectors. The charge controller confirmed successful charging through its display.
5. The 12 V supply terminal block and greenhouse system ground were connected to the load terminals of the charge controller.
6. The controller display indicated power transfer to the load. This was verified by measuring the potential difference across the load terminals with a multimeter.

These steps confirmed correct wiring and successful operation of the solar power subsystem as imaged in Appendix H. The charge controller additionally provides low-voltage protection by disconnecting the load if the battery voltage drops below a safe threshold. During full system testing, it was confirmed that the solar array supplied sufficient power to operate the greenhouse without discharging the battery to unsafe levels even during days with high cloud cover.

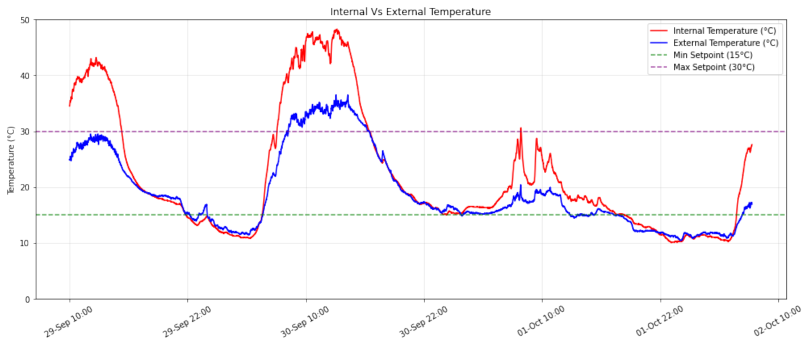
6.6 Enviromental Response

With all sensors, actuators, solar power components, and monitoring systems successfully tested, installed, and integrated, the complete greenhouse system can now be operated with all functions active. The system is powered on and allowed to continuously measure environmental conditions while autonomously regulating the internal climate toward the desired setpoints.

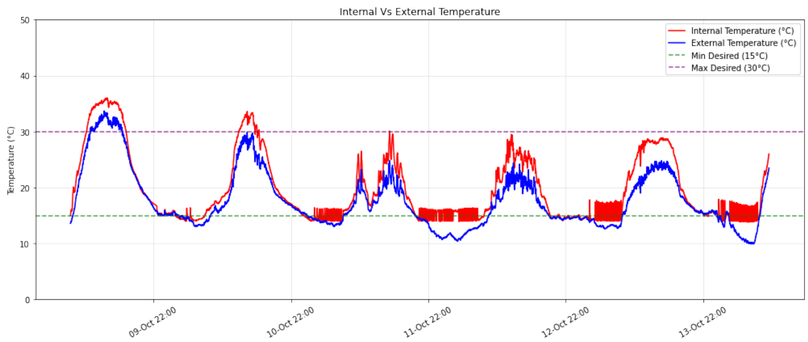
6.6.1 Dual Phase Test

To evaluate the performance of the greenhouse control system, a two-phase test procedure was conducted. In the first phase, the greenhouse was operated for three full day–night cycles with all sensors and monitoring systems active, but without the actuators connected. This allowed the internal environment to respond passively to external conditions, providing a baseline measurement of natural fluctuations in temperature, humidity, and soil moisture.

In the second phase, the actuators (ventilation, irrigation, and heating) were connected and enabled. The greenhouse was again run for more than three full day–night cycles, with the control system actively attempting to regulate conditions. This phase demonstrates the effectiveness of the system in stabilizing the greenhouse environment. The temperature, humidity and soil data from both phases were collected from Thingspeak and plotted using python, enabling a direct comparison between passive response and active control. The resulting figures highlight the extent to which the actuators improve environmental stability and overall system performance.



(a) Temperature profile with **Actuators off**



(b) Temperature profile with **Actuators on**

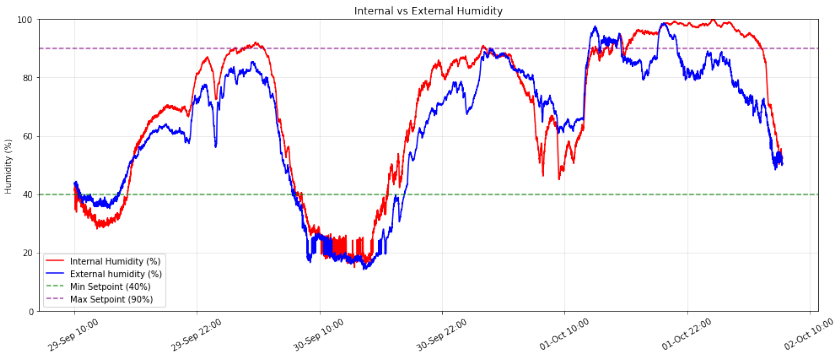
Figure 6.7: Internal vs. external temperature profiles with (a) Actuators off and (b) Actuators on.

Temperature

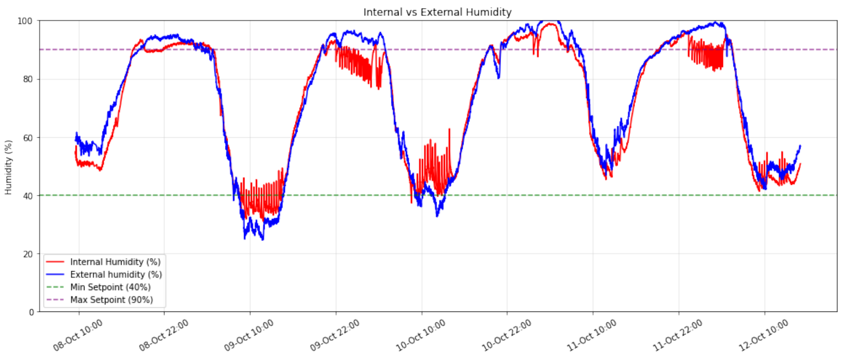
Without any active control, the internal temperature (shown in red) was allowed to fluctuate naturally with ambient conditions and rose to a peak of approximately 48 °C during the day, significantly exceeding optimal growing conditions. At night, the temperature dropped well below desired levels, reaching approximately 10 °C, highlighting the system's vulnerability to external temperature extremes under passive operation.

In contrast, during the **actuator-on** test plotted in figure 6.7b, the greenhouse exhibited an improved thermal response. The heating system successfully maintained the internal temperature above the desired setpoint of 15 °C during cooler periods, keeping the environment consistently warmer than outside conditions. However, under hot daytime conditions, the system faced challenges in maintaining internal temperatures below the 30 °C setpoint. Internal temperatures remained less than 5 °C above ambient. Despite this limitation, the active ventilation, intake fans, and misting system provided measurable cooling benefits compared to the uncontrolled scenario, in which temperatures rose to approximately 15 °C above ambient due to the unchecked greenhouse effect heating the internal environment.

Humidity



(a) Humidity profile with **Actuators off**



(b) Humidity profile with **Actuators on**

Figure 6.8: Internal vs. external humidity profiles (a) actuators off and (b) actuators on.

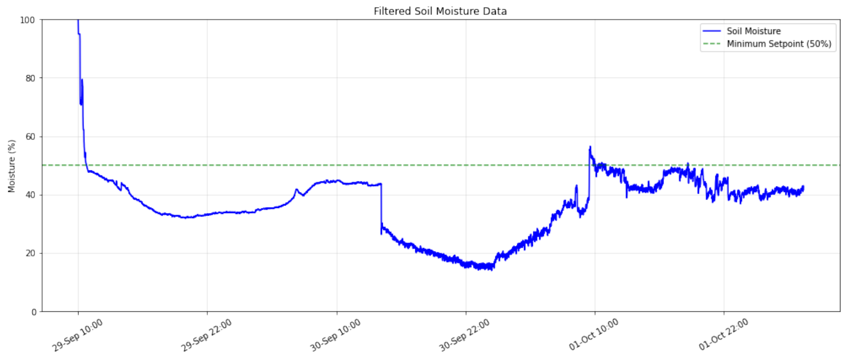
During the **actuator-off** phase, the internal and external humidity profiles closely mirrored one another, with relative humidity varying widely between 20% and 100% over the day–night cycle. This variation reflects the natural dependence of relative humidity on temperature, decreasing during the warmer daytime periods and increasing overnight.

During the **actuator-on** phase plotted in 6.8b, the system responded to low humidity by activating the misting function every 20 minutes, visible as periodic spikes in the recorded data. These misting periods effectively restored the internal humidity to above 40% (within the desired range), demonstrating successful environmental regulation. At night during cold periods, operation of the heater caused a decrease in humidity, an unintended but beneficial effect that maintained more balanced conditions during peak RH levels. However, at times when the heater was inactive due to adequate internal temperatures, the humidity occasionally exceeded the desired maximum setpoint.

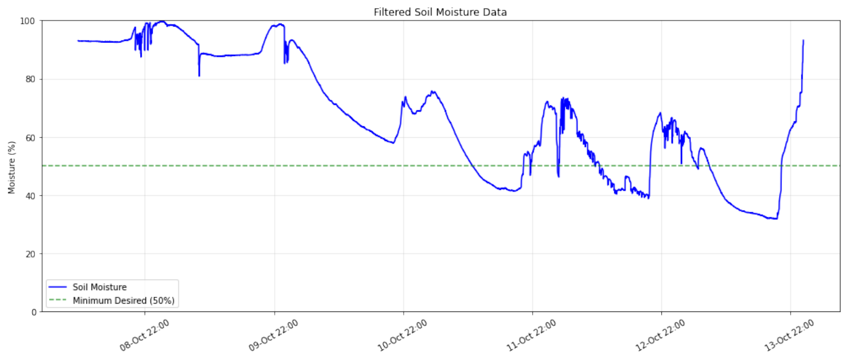
Overall, the actuator-on phase exhibited a more stable humidity range of 40%–100%, compared to the uncontrolled 20%–100% range. While the system was unable to maintain the desired maximum humidity limit of 90%, it successfully prevented humidity from dropping below the 40% minimum, demonstrating effective humidity control through intermittent misting.

Soil Moisture

During the initial **actuator off** phase, when the mister was inactive, the soil moisture content gradually declined and remained below the setpoint as the soil medium dried out. A slight increase was observed on the third day following a cold, rainy night, likely due to condensation re-entering the system. However soil moisture levels were consistently below the desired level.



(a) Soil Moisture profile with **Actuators off**



(b) Soil Moisture profile with **Actuators on**

Figure 6.9: Soil Moisture profile with (a) actuators off and (b) actuators on.

In contrast, during the **actuator on** test, the initial hot days prompted intermittent misting for evaporative cooling. Accordingly, the soil-moisture trace in Figure 6.9b remained well saturated for several days before declining to the setpoint, at which point the misting system activated to restore the desired level. During the subsequent cooler days, a recurring pattern emerged with nighttime moisture loss followed by controlled daytime misting events that raised levels back above the desired setpoint. On average, the system maintained soil moisture within, or slightly above the target range during the day, demonstrating effective, sensor-driven control.

The fluctuations and response delays observed, are attributed to the misting system's operational settings, restricted to daytime activation and governed by a duty cycle of two minutes on every twenty minutes, as described in Appendix C. These constraints were intentionally implemented to prevent overwatering, allow time for sensor feedback stabilization, and avoid unintended nighttime cooling when thermal regulation takes priority.

6.6.2 Light Observation and Supplementation

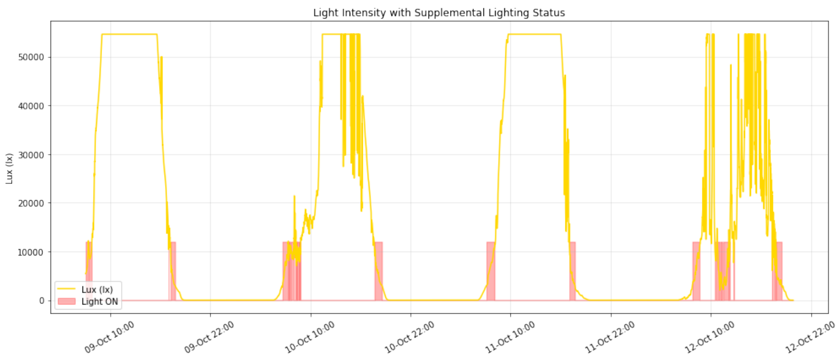


Figure 6.10: Lux Profile

Using Python for data processing and visualization, the plot in figure 6.10 illustrates the digital on/off state of the supplemental LED lighting alongside the measured light intensity trace. As calculated in Appendix G, the LED was designed to provide 200 $\mu mol{\cdot}m^{-2}{\cdot}s^{-1}$ PPFD and was configured to activate whenever the ambient sunlight levels dropped below this threshold, but only during daylight hours. This corresponds to an illuminance of approximately 10 800 lux using the conversion stipulated in Formula 2.1. In the above plot, the lighting can be observed switching on during the early morning and late evening periods, as well as intermittently on the fourth day when cloud cover reduced incoming solar radiation to below the desired PPFD levels. This behavior demonstrates successful automatic light supplementation, maintaining consistent photosynthetic photon levels during the day despite varying natural light conditions.

6.6.3 CO_2 Observation

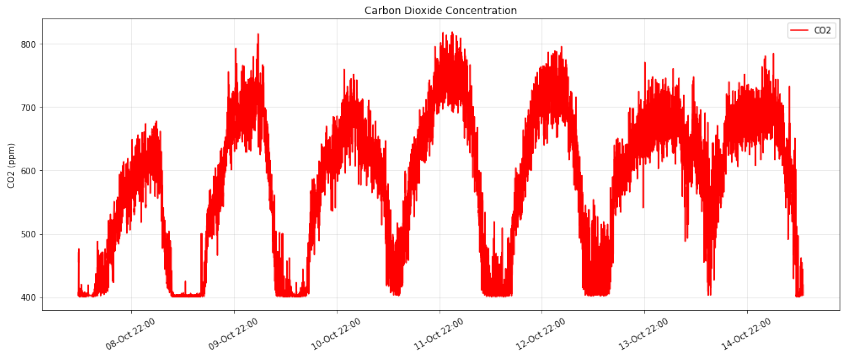


Figure 6.11: CO_2 Profile

The CO_2 profile provides valuable insight by highlighting a distinct day–night cycle, with concentrations rising during the night as a result of plant respiration. This trend is consistent with general ambient CO_2 behavior and does not indicate any problematic accumulation within the greenhouse. Since the levels remained within healthy ranges for plant growth, no active control or actuation was required, such as venting to regulate concentrations.

6.7 Evaluation Using Performance Requirements

Table 6.2: Performance Requirement Verification

PR Number	**Description**	**Target**	**Actual**	**Comment**
1	Resist Wind speeds	18.81 kph	Not Tested	Validated through calculations
2	Temperature sensing range	-10–50 °C	-40–80 °C	Passed
3	Soil moisture sensing	0–100 %	0–100 %	Passed
4	Humidity sensing	0–100 RH%	0–100 RH%	Passed
5	CO_2 sensing range	0–1000 ppm	0–5000 ppm	Passed
6	Light sensing	0–600 $\mu mol \cdot m^{-2} \cdot s^{-1}$	0–925 $\mu mol \cdot m^{-2} \cdot s^{-1}$	Passed
7	Latency in displaying environmental conditions	<120 s	~30 s average	Passed
8	Control temperature	15 min–30 max °C	15–35 °C maintained	15 min Passed — 30 max Failed

Continued on next page

PR Number	Description	Target	Actual	Comment
9	Control soil moisture content	50–100%	50–100% maintained during day	Passed
10	Control humidity	40% min–90% max	40–100% maintained	40% min Passed — 90% max Failed
11	Supplemental lighting intensity	200 $\mu mol \cdot m^{-2} \cdot s^{-1}$	Not tested	Validated in calculations
12	Energy consumption	0–1200 Wh	980 Wh over 24 h	Validated in calculations
13	Battery storage capacity	24 h	Not Tested	Remained sufficiently charged via solar over cloudy periods

6.8 Full system Test with Plants Growth

The operational success of the greenhouse system was validated through a controlled two-week growth trial of microgreens, conducted to verify the system's ability to maintain suitable environmental conditions for plant growth. The experiment served as a performance test against the design requirements related to temperature, humidity, and irrigation control. Four microgreen varieties (peas, radish, beetroot, and arugula) were planted within the greenhouse. As shown in the before-and-after photographs below, all varieties exhibited healthy germination and growth, confirming that the system effectively supported sustained plant development and met its intended functional objectives.

(a) Before: microgreen seeds at planting.

(b) After: successful microgreen growth.

Figure 6.12: Two-week microgreen growth trial conducted in the greenhouse.

Chapter 7

Conclusions

Future Reccomendations

The results indicated that the design was effective in providing reliable monitoring of greenhouse conditions, as well as control of water supply, humidity, and supplemental lighting. However, the system showed limitations in its ability to regulate higher temperatures during hotter days. For future development of an automated greenhouse in the warmer climate of South Africa, alternative cooling strategies could be explored, such as motorized shading or the integration of an evaporative cooling pad. In addition, the sizing of intake and exhaust fans could be optimized with airflow capacity in mind, favoring an overcompensated design to ensure adequate ventilation under peak heat conditions.

Conclusions

The design proved effective in monitoring all key parameters required to sustain healthy plant growth. Automated heating successfully maintained temperatures above the setpoint during cooler nights, while the misting system provided sufficient control of both humidity and water supply.

The vent and intake fans contributed to improved airflow, helping to limit temperature rise compared to conditions without these actuators. Although overheating remained a challenge, the greenhouse system demonstrated the ability to regulate a microclimate capable of supporting plant life, while providing the user with real-time and historical data and control through the wireless dashboards.

In addition, the solar power system enabled fully off-grid operation, allowing the greenhouse to run entirely on renewable energy. This not only reduced the system's carbon footprint but also highlighted its potential as a sustainable solution to address food insecurity by providing controlled growing conditions, with the primary limitation being high-temperature management.

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Appendix A: Concept Evaluation

This section evaluates and expands on the developed concepts. A weighted scoring system was used to determine the best design for each subsystem. Key criteria such as efficiency, cost, and manufacturability were assigned importance weights, and each concept was rated from 1–5. Scores were calculated by multiplying ratings by weights and summing the results. The highest-scoring concepts were selected to ensure optimal performance while aligning with project priorities. Although this quantitative method reduces bias, some subjectivity in the assessment remains.

A.1 Structure

Three structural options were considered to meet the small-scale greenhouse requirements. Criteria were weighted by importance and rated on a 1–5 scale. The weighting reflects project priorities: temperature management (30%) and cost effectiveness (30%) are most critical, while manufacturability (20%) and space efficiency (20%) are secondary considerations.

Criterion	**Weight (%)**	**Even Span**	**Lean-to**	**Arch**	**Rationale**
Temperature Management	30	3	3	3	All options allow moderate control; lean-to benefits from thermal mass of shared wall, but may overheat in summer.
Ease of Manufacturability	20	2	4	1	Lean-to can require fewer materials by leveraging an existing wall; even span is moderate; arch is most complex.
Space Efficiency & Footprint	20	2	3	2	Lean-to can optimize space against a building; others require more free-standing footprint.
Cost Effectiveness	30	2	3	1	Lean-to reduces material and covering costs; even span moderate; arch requires more structure and covering.
Weighted Total	**100**	**2.3**	**3.2**	**1.8**	Lean-to achieves the best balance across cost, space, and manufacturability.

Table A.1: Weighted Concept Evaluation of Greenhouse Structures

For small-scale applications, the lean-to option emerges as the most practical solution, balancing cost, space utilization, and manufacturability, while still offering adequate thermal performance.

A.2 Covering Material

The weighting was chosen to reflect the relative importance of each criterion in greenhouse performance. Durability (30%) was weighted highest as the covering must withstand weathering and mechanical stress. Light transmissibility (20%) and thermal insulation (20%) were both critical to plant growth. Cost effectiveness (20%) was included to balance performance with budget, while ease of manufacturability (10%) was given a smaller weight since fabrication complexity is less important than long-term functionality.

Criterion	**Weight (%)**	**Glass**	**Soft Plastic Covering**	**Perspex Acrylic**	**Rationale**
Light Transmissibility	20	4	3	3	Glass transmits light exceptionally well; soft plastic is less transparent; Perspex is good but typically slightly lower than glass.
Durability	30	2	1	4	Glass is brittle and prone to breakage; soft plastic tears/UV-degrades; Perspex is impact-resistant with good weathering.
Cost Effectiveness	20	3	5	2	Soft plastic has lowest upfront cost; glass moderate; Perspex is higher cost per m^2.
Ease of Manufacturability	10	1	2	3	Glass is heavy and difficult to cut/fit; soft plastic easier to handle; Perspex is lightweight and easy to cut/shape.
Thermal Insulation	20	3	1	3	Glass and Perspex provide reasonable insulation; thin soft plastic performs poorly.
Weighted Total	**100**	**2.7**	**2.3**	**3.1**	

Table A.2: Weighted Concept Evaluation of Greenhouse Covering Materials

Perspex was selected as the covering material due to its durability, ease of fabrication and ability to be cut to shape, and favourable balance between thermal insulation and light transmittance.

A.3 Active Heating

Temperature management and cost effectiveness were weighted highest at 30% each, as maintaining a stable climate and operating affordability are the most critical factors for long-term success.

Table A.3: Weighted Concept Evaluation of Heating Options

Criterion	**Weight (%)**	**Electric Heater**	**Gas Heater**	**Heat Lamp**	**Rationale**
Temperature Management	30	3	4	3	Gas provides strong heating; electric moderate; lamps uneven.
Ease of Manufacturability	20	4	1	2	Electric is plug-and-play; gas requires plumbing; lamps simple but less effective.
Off-Grid Capabilities	20	4	1	1	Electric integrates with solar; gas not sustainable; lamps not suited for solar.
Cost Effectiveness	30	2	3	1	Gas fuel is cheaper; electric moderate; lamps inefficient per area.
Weighted Total	100	**3.1**	**2.5**	**2.2**	

An electric heater was selected as it offers efficient, evenly distributed heating, seamless integration with the solar power system, and lower cost and complexity compared to gas heaters or heat lamps.

A.4 Active Cooling

The weighting was selected to reflect the relative importance of cooling performance in maintaining plant health. Temperature management (30%) was given the highest weight as this is the primary purpose of cooling. Cost effectiveness (30%) was also heavily weighted due to budget constraints for long-term operation. Off-grid capabilities (20%) were prioritized to ensure compatibility with renewable systems. Finally, ease of manufacturability (20%) was included to account for installation complexity and practicality.

Criterion	**Weight (%)**	**Actuated Ventilation**	**Evaporative Cooling Module**	**Misting**	**Rationale**
Temperature Management	30	4	2	3	Ventilation exchanges hot air for ambient; evaporative modules underperform in humidity; misting adds moderate evaporative cooling.
Ease of Manufacturability	20	4	1	3	Fans + simple controls vs. pumps, pads, and ducting; misting needs basic plumbing and nozzle layout.
Off-Grid Capabilities	20	3	2	2	DC/efficient fans suit solar; evaporative modules draw more power; misting depends on reliable pressurized water.
Cost Effectiveness	30	3	1	3	Ventilation: moderate ; evaporative: high cost and energy; misting: low cost but ongoing water use.
Weighted Total	**100**	**3.5**	**1.5**	**2.8**	

Table A.4: Weighted Concept Evaluation of Cooling Options

Actuated ventilation emerges as the most economical and energy-efficient cooling method, as it requires only fans to exchange hot and cool air. Its drawback is that it can only cool the greenhouse down to the ambient temperature outside. To enhance this, active ventilation can be combined with misting. Misting promotes evaporative cooling within the greenhouse and, although it requires a pressurized water supply and humidity control, it significantly improves temperature regulation. In contrast, evaporative cooling modules are less favorable due to their high cost, greater power demand, and limited performance in humid environments. Therefore, a combined ventilation and misting approach is the most balanced solution, addressing both cost-effectiveness and cooling performance.

A.5 Irrigation

The weighting was selected to reflect the importance of reliable soil moisture control (30%) and cost effectiveness (30%) as primary design drivers. Coupled environment control capabilities (15%) were included to account for irrigation methods that are able to also influence temperature and humidity. Ease of manufacturability (15%) was considered for practicality, while off-grid capabilities (10%) were given a smaller weight as irrigation can often be supported by external water supplies.

Criterion	Weight (%)	Drip Fed	Ebb & Flow	Misting	Rationale
Soil Moisture Control	30	3	3	2	Drip and ebb-flow target roots well; misting less precise for soil.
Ease of Manufacturability	15	4	1	4	Drip/mist easy to set up; ebb-flow complex with pumps/reservoirs.
Off-Grid Capabilities	10	3	2	1	Drip can run on solar pumps; ebb-flow requires circulation; misting needs high pressure.
Coupled Env. Control	15	0	0	5	Only misting can also improve humidity/temperature Control.
Cost Effectiveness	30	4	2	4	Drip and misting affordable; ebb-flow expensive/complex.
Weighted Total	**100**	**3.00**	**1.85**	**3.25**	

Table A.5: Weighted Concept Evaluation of Irrigation Methods

Misting emerges as the most effective method due to its dual role in both irrigation and environmental control, despite requiring higher water pressure. A misting system with sensor-based control is therefore preferred for optimizing both plant growth and climate regulation.

A.6 Controller Selection

The weights were assigned based on the relative importance of each criterion to the concept: GPIO flexibility (30%) is most critical for hardware interfacing, programmability and cost each carry significant weight (20% each), while WiFi (20%) and power consumption (10%) are also key but secondary factors. PLCs were considered but ruled out early due to their high cost, complexity, and limited suitability for rapid prototyping or application in small-scale systems.

Criterion	**Weight (%)**	**Raspberry Pi 4B**	**ESP32**	**Arduino Uno**	**Rationale**
Input/Output GPIO Flexibility	30	3	4	3	ESP32 has more versatile GPIO including analog inputs; Pi and Arduino provide sufficient but less flexible options.
Ease of Programmability	20	4	4	4	All three have strong community support and beginner-friendly development environments.
WiFi Capabilities	20	4	4	2	ESP32 and Pi include WiFi by default; Arduino requires shields/modules, increasing cost and complexity.
Power Consumption	10	4	3	3	Raspberry Pi is relatively efficient; ESP32 moderate; Arduino low but limited in capability.
Cost Effectiveness	20	2	4	3	ESP32 offers the best feature-to-price ratio; Arduino is low-cost but less capable; Raspberry Pi is more expensive.
Weighted Total	**100**	**3.3**	**3.9**	**3.0**	ESP32 emerges as the most balanced choice overall.

Table A.6: Weighted Evaluation of Microcontroller Options

The ESP32 Offers a balance of performance and cost, with built-in Wi-Fi and Bluetooth, making it ideal for IoT and wireless applications.

Appendix B: Physical Decomposition

The following sample physical decomposition figures break down the physical components necessary in the primary subsystems. The use of a system engineering approach delves into physical decomposition of the main subsystems at higher detail (namely Environmental Monitoring and Actuation) and are numbered 1 through 2.2 as numbered in the functional decomposition in figure 3.1.

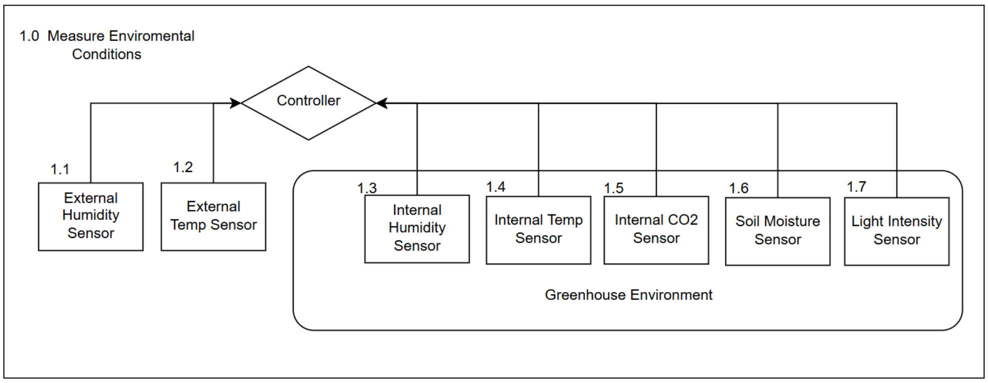


Figure B.1: Environmental monitoring Physical Decomposition

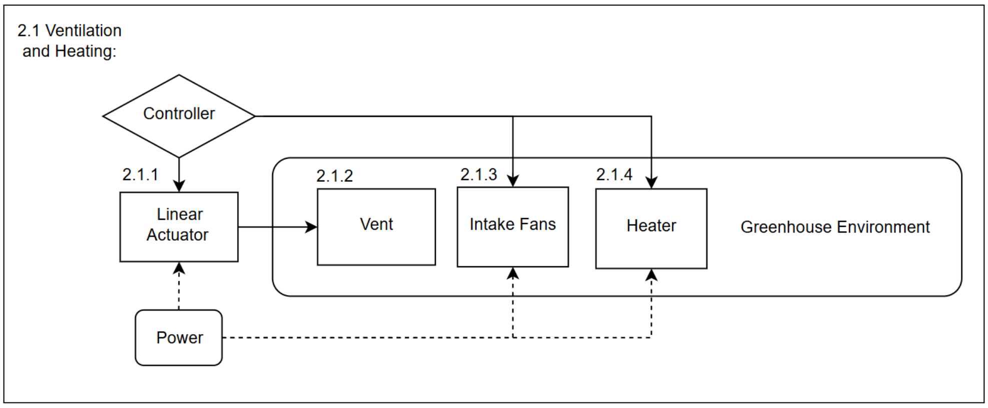


Figure B.2: Ventilation and Heating Actuation Physical Decomposition

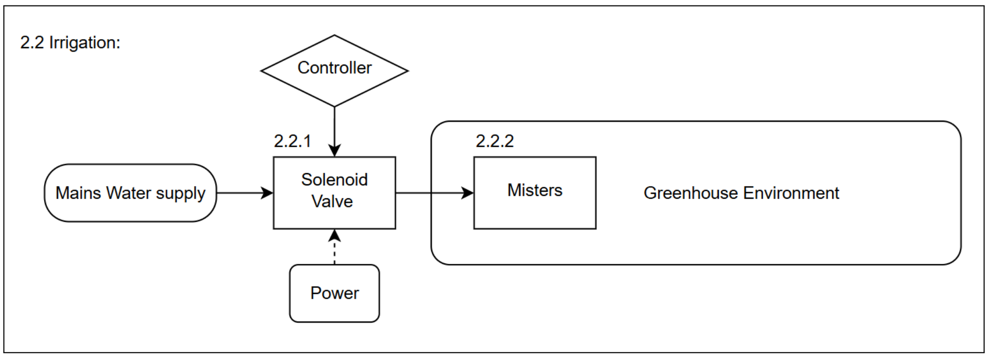


Figure B.3: Irrigation Physical Decomposition

Appendix C: Sample Software Control Algorithms

C.1 Temperature Control System

Temperature control is given the highest priority among all environmental parameters. The system continuously measures the internal temperature of the greenhouse and compares the readings to user-defined setpoints. When the temperature falls outside the desired range, the appropriate actuators namely the vent, fans, heater, or misters are activated or deactivated accordingly.

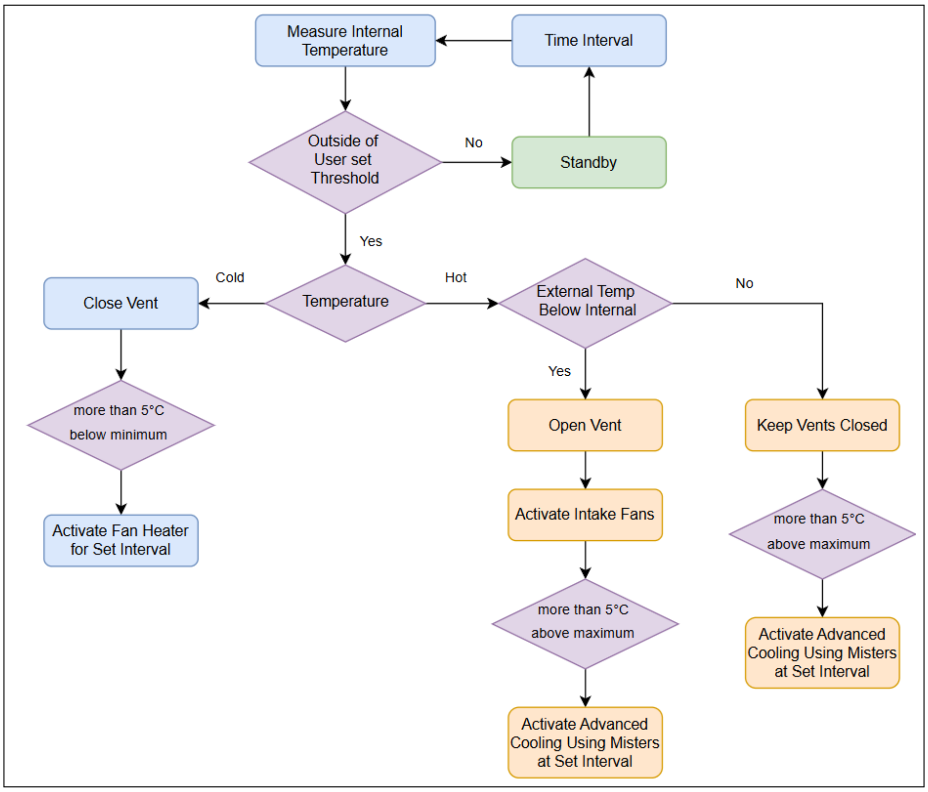


Figure C.1: Temperature control

To prevent excessive watering, the misting system is only enabled for thermal regulation when the internal temperature exceeds the upper setpoint by at least 5 °C This is designated as 'Advanced Cooling'. This allows passive evaporative cooling to occur without over-saturating the soil. All temperature thresholds are implemented with hysteresis logic to prevent unnecessary switching and ensure stable control behavior.

C.2 Airflow Control System

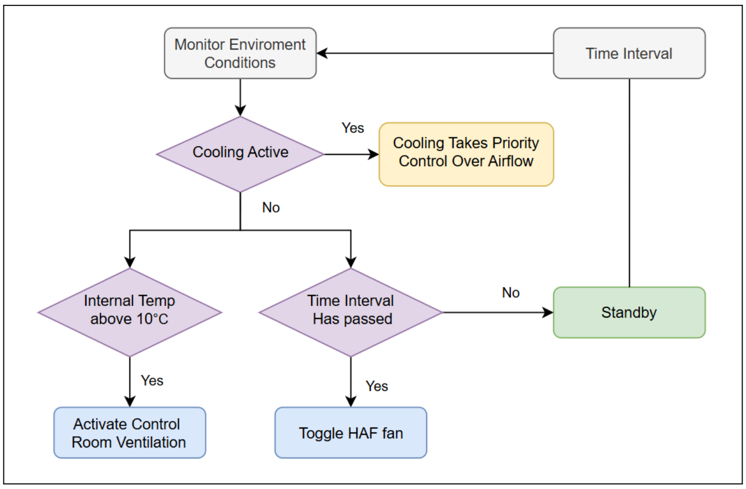


Figure C.2: Airflow control

Airflow control is implemented according to the flowchart shown above. While temperature regulation takes priority over airflow operations, the airflow subsystem functions independently when cooling is not required. In such cases, the dedicated control room fan is activated whenever the ambient temperature inside the greenhouse exceeds 10 °C, ensuring adequate ventilation and preventing electrical components from overheating. Additionaly when temperature control is inactive, the external HAF/intake fan operates intermittently to maintain healthy air circulation and fulfill horizontal-airflow (HAF) requirements within the greenhouse.

C.3 Irrigation Control System

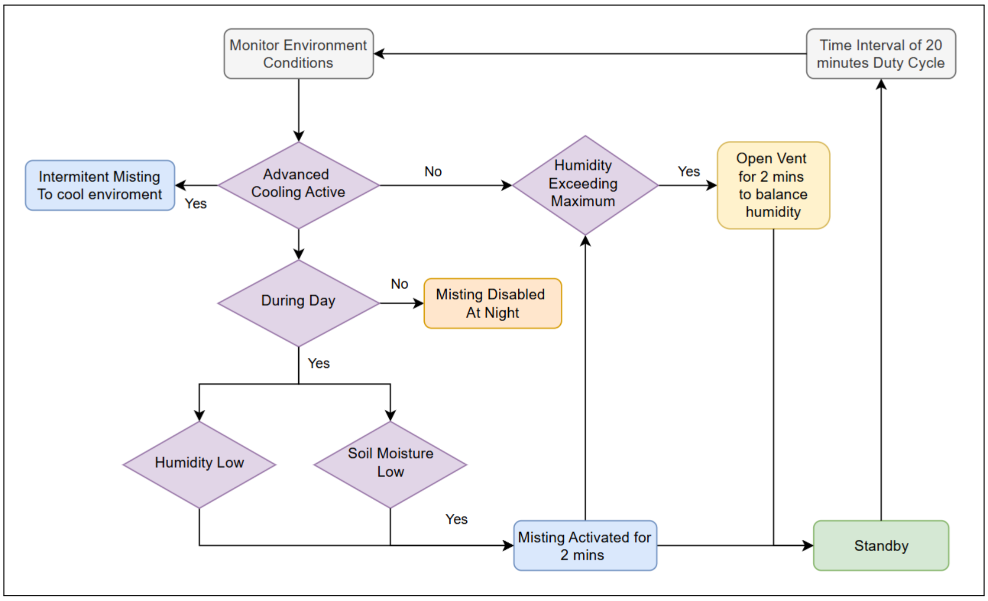


Figure C.3: Irrigation control

The irrigation control flowchart outlines the misting logic implemented in the system. The controller first determines whether 'advanced cooling' is required, in such cases temperature control takes priority, and the misters are activated intermittently to promote passive evaporative cooling of the environment. When advanced cooling is not active, the system compares humidity and soil moisture levels against their respective setpoints. If either parameter falls below the desired range, misting is initiated. Misting is disabled during nighttime hours to maintain lower humidity levels, reducing the risk of fungal growth, and to prevent unwanted cooling effects that could lower temperatures below optimal levels. Hysteresis control is implemented in all misting decisions to prevent rapid cycling and ensure stable operation.

All misting operations are regulated by an overall duty cycle, allowing the misters to run for two minutes followed by a twenty-minute pause. This prevents overwatering and ensures sufficient time for sensor feedback to stabilize. Additionally, if humidity levels exceed the maximum threshold after misting, the ventilation system is temporarily activated to restore balanced humidity conditions within the greenhouse.

C.4 Light Control System

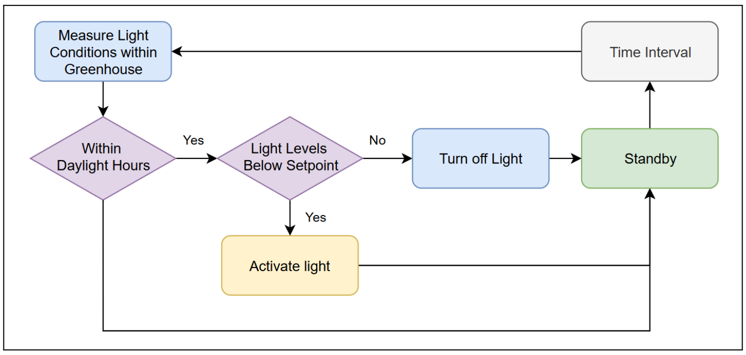


Figure C.4: Light control

The light control flowchart outlines the logic governing supplemental lighting within the system. Supplemental lighting is restricted to operate only during daylight hours. When the measured light intensity falls below the desired setpoint of 200 $\mu mol \cdot m^{-2} \cdot s^{-1}$, the system activates the grow lights to maintain adequate illumination for plant growth. Hysteresis logic is incorporated to prevent unnecessary on–off cycling, ensuring stable operation and reducing wear on the lighting components.

Appendix D: Techno-economic Analysis

This appendix aims to discuss the project in terms of its technical and economic impact, various factors such as planning procedures, budget and ROI shall be presented.

D.1 Budget

Proposed Budget:

Activity	Engineering Time		Running Costs	Facility Use	Capital Costs	MMW			TOTAL
						Labour		Material	
	hr	R	R	R	R	hr	R	R	R
Conduct Research	65	29250	500	0	0	0	0	0	29750
Composing design requirements	30	13500	0	0	0	0	0	0	13500
Choose a design methodology	10	4500	0	0	0	0	0	0	4500
Conceptual system design	80	36000	0	0	0	0	0	0	36000
Design review	70	31500	0	0	0	0	0	0	31500
Component selection	25	11250	0	0	3000	0	0	0	14250
Electrical and solar design	25	11250	0	0	2000	10	3000	2000	18250
Control development	30	13500	200	0	0	0	0	0	13700
Construction	30	13500	500	500		25	7500	2000	24000
Subsystem integration	20	9000		0	0	0	0	0	9000
Testing and calibration	35	15750	500	500	0	0	0	0	16750
Data logging and analysis	40	18000	500	500	0	0	0	0	19000
Documentation and Final Report	60	27000	100	0	0	0	0	0	27100
TOTAL	520	234000	2300	1500	5000	35	10500	4000	**257300**

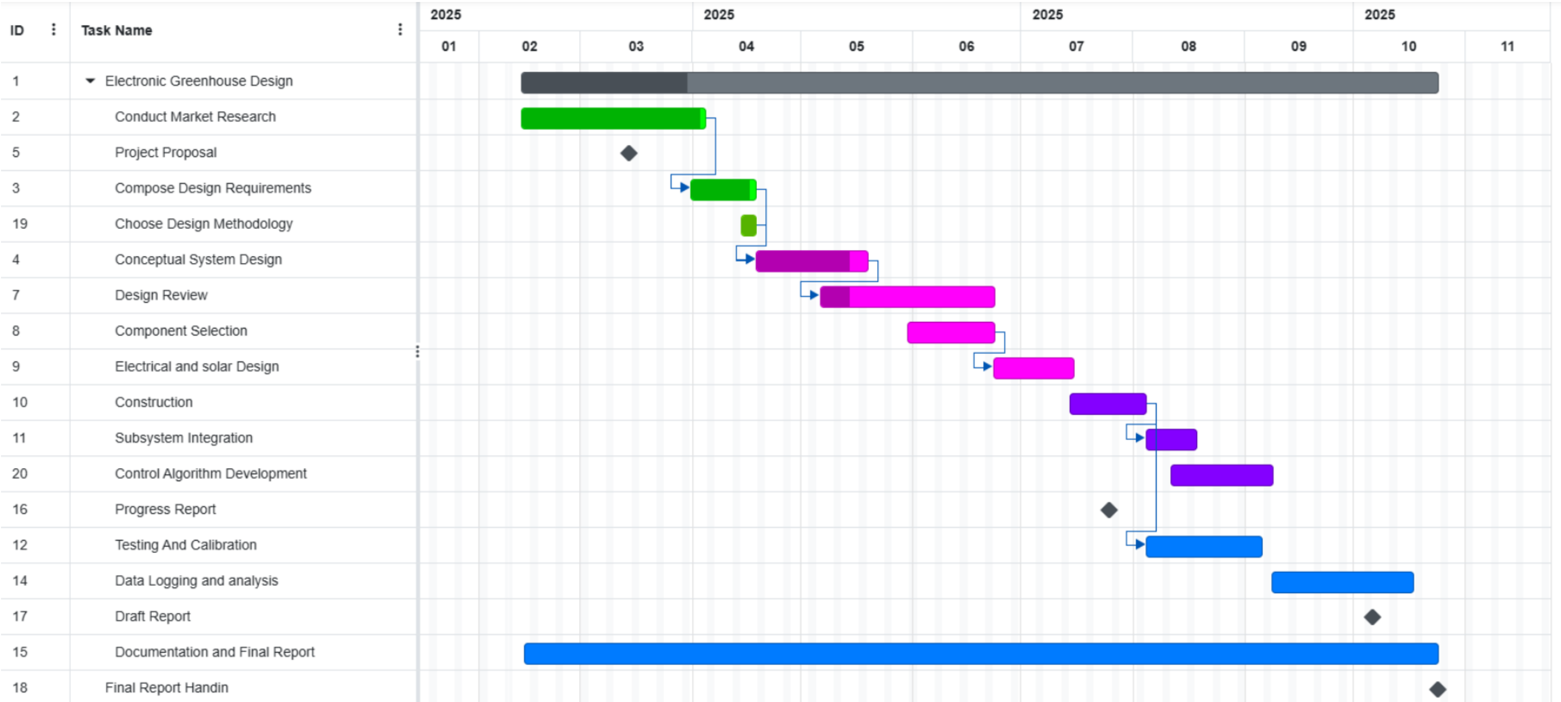
Actual Costs:

Activity	Engineering Time		Running Costs	Facility Use	Capital Costs	MMW			TOTAL
						Labour		Material	
	hr	R	R	R	R	hr	R	R	R
Conduct Research	65	29250	0	0	0	0	0	0	29250
Composing design requirements	30	13500	0	0	0	0	0	0	13500
Choose a design methodology	10	4500	0	0	0	0	0	0	4500
Conceptual system design	80	36000	0	0	0	0	0	0	36000
Design review	70	31500	0	0	0	0	0	0	31500
Component selection	25	11250	0	0	4674.84	0	0	0	15924.84
Electrical and solar design	25	11250	0	0	0	0	0		11250
Control development	30	13500	169.88	0	0	0	0	0	13669.88
Construction	90	40500	509.69	500	0	29.5	8850	849.95	51209.64
Subsystem integration	35	15750	198.21	0	0	0	0	0	15948.21
Testing and calibration	80	36000	33	0	0	0	0	0	36033
Data logging and analysis	30	13500	22	0	0	0	0	0	13522
Documentation and Final Report	60	27000	0	0	0	0	0	0	27000
TOTAL	630	283500	932.78125	500	4674.84	29.5	8850	849.95	299307.6

Figure D.1: Proposed Project Budget and Actual Costs

The primary deviation between the planned and actual budget resulted from additional engineering time required during the construction and testing phases, which took longer than anticipated. In contrast, material and labour costs remained close to the projected values. The low manufacturing cost was largely due to the use of standard aluminium extrusions readily available in the workshop. The Perspex panels and sensors accounted for most of the remaining capital expenses, their costs were still slightly lower than originally estimated in the proposed budget.

D.2 Planning and Time Management



The project adhered closely to the planned timeline, with the main deviation occurring during the physical construction phase. Securing and sealing the greenhouse panels and constructing the control room along with wiring proved more time-consuming than anticipated, causing a temporary backlog. This was resolved by late September, allowing several weeks for the crucial data logging, analysis, and control logic fine-tuning needed based on system feedback.

D.3 Technical Impact

The project demonstrates the feasibility of a cost-effective, automated greenhouse system that minimizes manual labour and maintenance through intelligent control and wireless monitoring. The design integrates sensors, actuators, and a microcontroller-based control system to automatically regulate key environmental variables such as temperature and humidity.

Experimental testing validated the performance of the Perspex greenhouse equipped with actuated ventilation and misting systems, both contributing to stable internal climate conditions. The system operates off-grid through a solar power supply, reinforcing its sustainability and suitability for remote or resource-limited areas. Overall, the project establishes a foundation for scalable, low-maintenance greenhouse automation, supporting sustainable food production for users with limited time, space, or access to conventional infrastructure.

D.4 Return on Investement

The return on investment (ROI) for the automated greenhouse system can be realized through the reduced operating costs and increased crop yield stability. By utilizing a solar-powered energy system, the design offsets the electricity expenses typically associated with environmental control systems. The greenhouses automation minimizes manual labour requirements, lowering maintenance and supervision costs over time. Although the initial setup cost is higher due to the inclusion of sensors, actuators, and control electronics, these expenses are recovered through long-term operational savings and improved production efficiency. The integration of renewable energy and intelligent control thus ensures a sustainable payback period.

D.5 Potential for Commercialization

Although the small size of the greenhouse limits its use for large commercial farming, the control methods and automation developed in this project could easily be scaled for larger greenhouses. The design uses standard and easily available parts, making it simple and cost-effective to reproduce. This gives it good potential for commercialization as a small-scale product aimed at communities, schools, or home growers. The use of solar power and automated control also makes it appealing as an educational and sustainable solution that promotes awareness of smart farming technologies.

Appendix E: Safety and Mitigation Procedures

This section outlines and summarizes the safety report in accordance with the laboratory safety considerations and potential risks applicable during the construction, testing, and operation phases of the project .

E.1 Safety

The project involves the development, assembly, and testing of a small-scale electronic greenhouse system that monitors and controls environmental parameters such as humidity, temperature, light, and water supply. The system makes use of rooftop infrastructure on the M&M engineering departments solar rooftop laboratory including mains water, solar radiation, and at times grid electricity for component power supply.

E.1.1 Type of Work Involved

- Electrical wiring (5V and 12V rails)
- Sensor installation (temperature, humidity, soil moisture)
- Microcontroller programming and deployment
- Use of water lines for irrigation systems
- Mechanical assembly of the greenhouse frame and Perspex panels

E.1.2 General Lab and Rooftop Safety

1. Work within the lab may only be conducted during allocated hours.
2. Closed shoes must always be worn.
3. No loose clothing may be worn.
4. Personal protective equipment (PPE) should be worn.
5. Good housekeeping practices must be maintained during construction, testing, and operation.
6. Extreme caution should be taken when working on the sunroof, as the lab houses other projects containing sensitive components.
7. Safety harnesses must be used when working near roof edges or elevated positions.
8. The safety report must be visible and accessible while working in the lab.
9. During operation, ensure that spectators remain within designated safety areas.

10. Tools and the greenhouse structure must not block roof drains, vents, or access pathways.
11. Maintain a clear 1-meter perimeter around other ongoing rooftop activities.

E.1.3 Electrical Safety

- All wiring must be insulated, with connections clearly labeled and weather-protected.
- Use fused circuits and DC buck converters for voltage regulation (e.g., 12V to 5V).
- Only low-voltage systems (≤24V DC) will be operated.
- Grid electricity access points must be clearly labeled, and all plugs and sockets must be waterproof.
- An emergency stop circuit to disconnect all power sources is mandatory.
- Electrical systems must be powered off before performing any maintenance or component adjustments.
- Cables leading to and from the greenhouse must not create tripping hazards.

E.1.4 Water Usage and Safety

- Avoid activating the irrigation system when electrical components are exposed.
- All water connections must use leak-proof tubing and connectors.
- Ensure proper drainage on the roof to avoid slip hazards or standing water near equipment.

E.1.5 Housekeeping and Shutdown

- Remove and return all tools and hardware at the end of each session.
- Sweep up debris, zip-tie cables, and label wiring.
- Disconnect all power and water connections.
- Report damaged components or safety hazards immediately.

E.1.6 Working at Heights

The working-at-heights safety training was completed on the 2nd of April 2025 by the student. To ensure the safety of all persons involved in the project, proper working-at-heights precautions must always be observed.

E.2 Risk Analysis and Mitigation Procedures

Table E.1: Activity-Based Risk Assessment

Activity	Risk	Risk Level	Risk Type	Mitigating Steps
Navigating lab	Tripping on loose cable or equipment	Low	P/E	Don't leave cables lying around
Work taking place on roof	Falling	Low	P	Use harness, avoid edges, complete heights training
Turning on/off equipment	Electric shock	Low	P	Check cables to ensure that insulation is intact, include warning signage
Use of hot glue gun	Accidentally contacting hot glue causing burn to skin	Medium	P	Wear protective gloves, ensure glue has enough time to cool before handling
Wiring of ports	Incorrect wiring may cause damage to components	Low	E	Make use of proper wiring diagram, label wires physically as laid out in diagram
Soldering connections of wires	Accidentally contact with the soldering iron can result in burns to the user or damage components	Medium	P/E	Keep a clean soldering workstation and use a solder stand. Turn off iron when not in use
Handling fluids near the system and operation of irrigation system	Spilling fluids can cause damage to electrical components	High	E	Do not bring fluids close to the system. Ensure irrigation system is sealed
Wiring of components	Electrical shock due to bad isolation of wires	Medium	P/E	Check all wires are well insulated before supplying power and ensure emergency stop is accessible. Include warning signage
Vent actuation	Hands in the way of moving parts causing potential small cut, bruise or pinch	Low	P	Keep hands clear of vent when operational
HAF/intake fan operation	Hands in the way of moving parts causing potential small cut, bruise or pinch	Low	P	Keep hands clear of fans when operational. Install safety mesh and coverings to prevent contact

Appendix F: Engineering Drawing

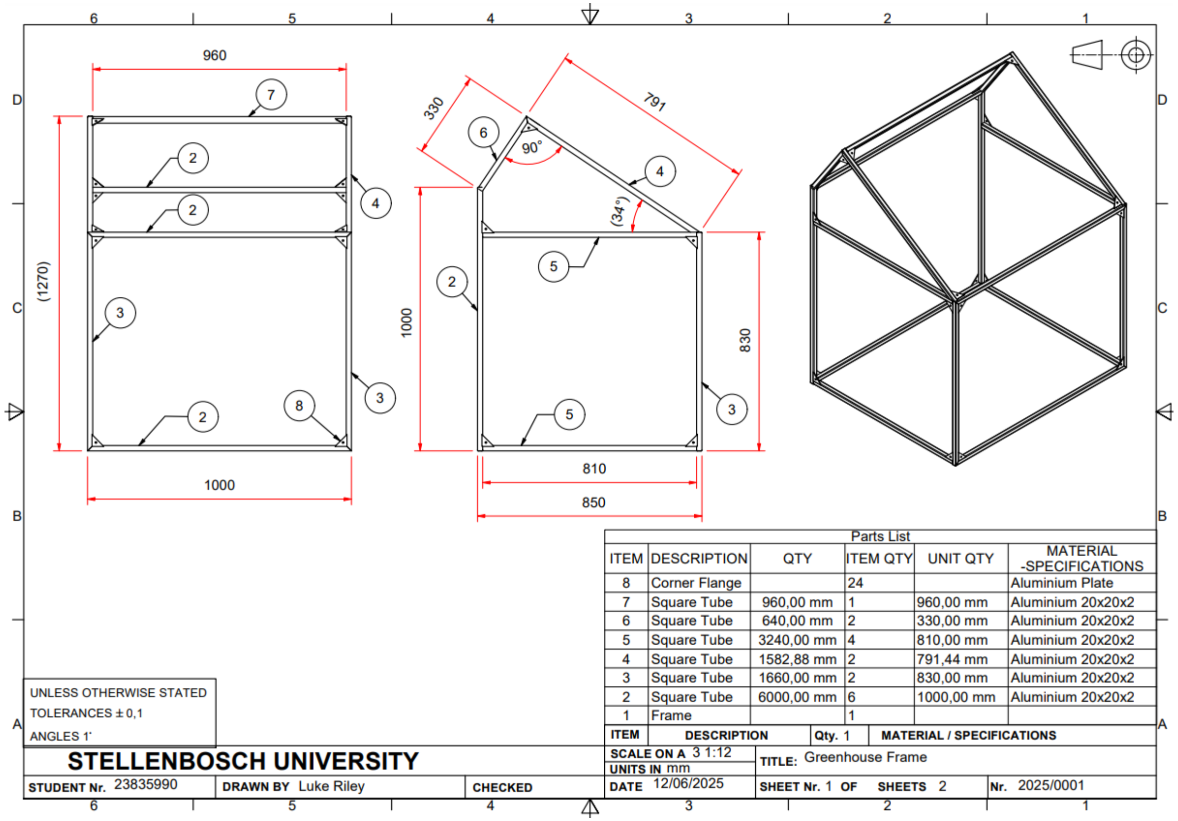


Figure F.1: Engineering Drawing

Appendix G: Sample Calculations

Sample Calculations:

1. Heat loss:

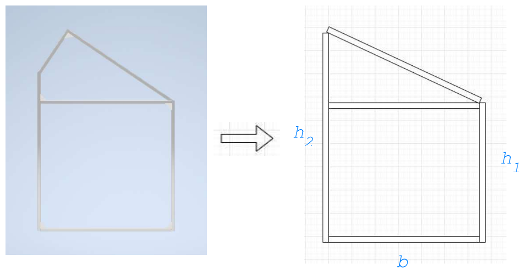
$h_1 := 820\ \mathrm{mm}$ $\quad h_2 := 1000\ \mathrm{mm}$ height of back panel and front panel

$l := 1000\ \mathrm{mm}$ $\quad b := 850\ \mathrm{mm}$ length and base of greenhouse foundation

$t := 0{,}003\ \mathrm{m}$ thickness of panels

The shape of the side profile of the greenhouse has been aproximated to simplify the calculations.

Asume ground surface area to not culminate any heat loss, but rather supply thermal mass to stabilize heat differences.



$$A := l \cdot h_1 \cdot 2 + l \cdot h_2 + \left(2 \cdot b \cdot \frac{\left(h_2 - h_1\right)}{2} + 2 \cdot b \cdot h_1\right)$$ area of all sides of perspex

$$A = 4{,}187\ \mathrm{m}^2$$

$$U := 5{,}7\ \frac{\mathrm{W}}{\mathrm{m}^2\ \mathrm{K}}$$ for 3mm thick perspex from Maizey plastics datasheet

$$\Delta T := 5\ \mathrm{K}$$ for a temperature difference of 5°C

$$Q := A \cdot U \cdot \Delta T$$ Heat Loss (1.1)

$$Q = 119{,}3295\ \mathrm{W}$$

$$Daily := Q \cdot 4\ \mathrm{hr}$$ asuming 4 hours of peak cold temperature differnece of 5°C during the night

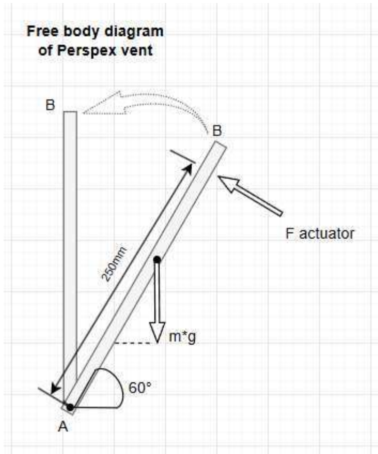
$$Daily = 0{,}4773\ \mathrm{kW\ hr}$$

$$\Delta T_{coldest} := (15 - 0)\ \mathrm{K}$$ asumming outside temperatures can reach 0°C at night and minimum setpoint temperature is 15°C

$$Q_{coldest} := A \cdot U \cdot \Delta T_{coldest}$$ (1.2)

$$Q_{coldest} = 357{,}9885\ \mathrm{W}$$

2. Vent Actuation:



$\rho_{perspex} := 1{,}19 \frac{g}{cm^3}$ density of perspex

$w_{vent} := 270\ mm$ width of vent

$m_{vent} := l \cdot w_{vent} \cdot t \cdot \rho_{perspex} = 0{,}9639\ kg$ mass of vent

$\theta := 60$ angle of incliniation of vent

$F_{weight} := m_{vent} \cdot 9{,}81 \frac{m}{s^2} = 9{,}4559\ N$ weight of vent acting at centroid

$\Sigma M_A := 0$

$M_{weight} := \left(-F_{weight}\right) \cdot \frac{w_{vent}}{2} \cdot \cos(\theta) = 1{,}2158\ N\ m$ moment around point A due to weight

$F_{actuator} := \frac{M_{weight}}{0{,}25\ m} = 4{,}8632\ N$ asuming that the linear actuator is connected to the panel at 250mm from the hinges at A and that it is initially acting perpindicularly to the panel

3. Cooling Power:

$\rho := 1{,}2 \frac{kg}{m^3}$ Air desnity a 50 CFM small intake fan aproximates to 0.02359m^3/s

$C := 0{,}6$ Discharge coefficient

$A_{vent} := 0{,}27\ m^2$ Area of vent

$g := 9{,}81 \frac{m}{s^2}$

$H := 1{,}2\ m$ Height difference between intake and vent

$T := 293\ K$ Average absolute Temperature assumed to be 20°C

$V_{vent} := C \cdot A_{vent} \cdot \sqrt{2 \cdot g \cdot H \cdot \frac{\Delta T}{T}}$ Bouyancy driven ventilation estimation using stack effect formula (3.1)

$V_{vent} = 0{,}1027 \frac{m^3}{s}$

$V_{fans} := 0{,}05 \frac{m^3}{s}$ Volumetric flowrate of two intake fans

$c_p := 1005 \frac{J}{kg\ K}$ Heat capacity of air

$\alpha := \frac{c_p \cdot \rho}{3600} = 0{,}335 \frac{Pa}{K}$ This approximation is commonly used in HVAC calculations to estimate the sensible heat removed through ventilation. (3.2)

$Q_{cool} := \alpha \cdot \left(V_{vent} + V_{fans}\right) \cdot \Delta T$ Sensible cooling load due to airflow (3.3)

$Q_{cool} = 0{,}2557\ W$ Removing ≈ 0.25 joules of heat per second

4. Lighting:

$PPFD_{required} := 400\ \text{micron}\ \frac{\text{mol}}{\text{m}^2\ \text{s}}$ to supply a range of needs of vegativive growth from seedlings to leafy greens

$PPFD_{sunlight} := 200\ \text{micron}\ \frac{\text{mol}}{\text{m}^2\ \text{s}}$ light intensity of sunlight averaged over 12 hour period asuming sunlight averages 200μmol/m2/s

$PPFD_{LED} := PPFD_{required} - PPFD_{sunlight} = 200\ \text{micron}\ \frac{\text{mol}}{\text{m}^2\ \text{s}}$ PPFD needed to match required level

$Grow\_Area := 850\ \text{mm} \cdot 850\ \text{mm}$ estimated maximum plant coverage within greenhouse

$Light\_Efficiency := 1,8\ \text{micron}\ \frac{\text{mol}}{\text{J}}$ micromols of PAR produced per Joule of electricity consumed asuming efficiency of LED to be 1,8 μmol/J

$Power_{LED} := PPFD_{LED} \cdot \frac{Grow\_Area}{Light\_Efficiency} = 80,2778\ \text{W}$ LED power required to supplement lighting (4)

5. Perspex Wind Loading:

$t := 3\ \text{mm}$ Plate thickness

$\sigma_m := 72\ \text{MPa}$ Tensile strength of Plexiglass from datasheet sourced through Maizey Plastics

$v := 0,35$ Poisons ratio

$b = 0,85\ \text{m}$

$E_{perspex} := 3,2\ \text{GPa}$

$v_{wind} := 40\ \frac{\text{km}}{\text{hr}}$ Maximum gust winds asumed to be up to 40km/h and acting normal to panel

$D := \frac{t^3 \cdot E_{perspex}}{12 \cdot \left(1 - v^2\right)} = 8,2051\ \text{N m}$ Flexural Rigidity of the Perspex plate asuming that it is uniformally loaded and simply supported at all ends (5.1)

$P := 0,5 \cdot \rho \cdot v_{wind}^{\ 2} = 74,0741\ \text{Pa}$ Dynamic pressure (5.2)

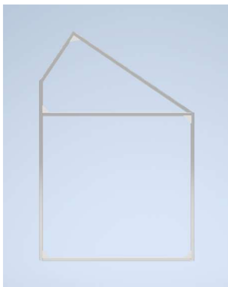
$y_{pmax} := P \cdot \frac{b^4}{D \cdot 64} = 0,0736\ \text{m}$ Maximum deflection in a uniformly loaded simply supported rectangular plate using Kirchhoff-Love and Mindlin-Reissner Plate theory (5.3)

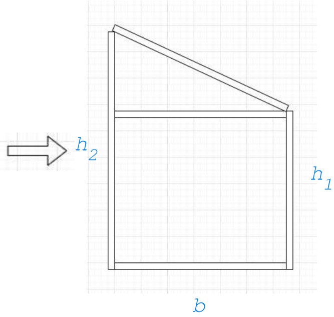
$\sigma_{pmax} := \frac{6 \cdot P \cdot b^2}{t^2} = 35,679\ \text{MPa}$ Maximum bending stress in a uniformly loaded simply supported rectangular plate using Kirchhoff-Love and Mindlin-Reissner Plate theory (5.4)

Tensile strength of perspex ≈ 70 MPa

$S_{perspex} := \frac{\sigma_m}{\sigma_{pmax}} = 2,018$ Factor of safety for perspex due to optimistic wind loading at 40km/h (5.5)

6. Aluminium Frame Structure:





$I := 4,92 \cdot 10^{-9}\ \mathrm{m}^4$ Moment of inertia

$c := 10\ \mathrm{mm}$ Distance from centroid to end of aluminium cross section

$E := 69\ \mathrm{GPa}$ Modulus of elasticity from Macsteel

$\sigma_y := 310\ \mathrm{MPa}$ Aluminium yield strength

$C_d := 1,2$ Drag coeffecient

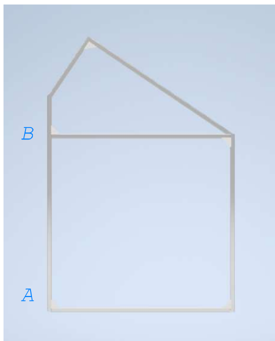
$A_{side} := b \cdot h_1 + b \cdot \frac{h_2 - h_1}{2}$ Area of side panel

$A_{side} = 0,7735\ \mathrm{m}^2$

$A_{section} := 80\ \mathrm{mm}^2$ Area of aliminium square tubing section

$v_{wind} := 40\ \frac{\mathrm{km}}{\mathrm{hr}}$ Maximum gust winds asumed to be up to 40km/h

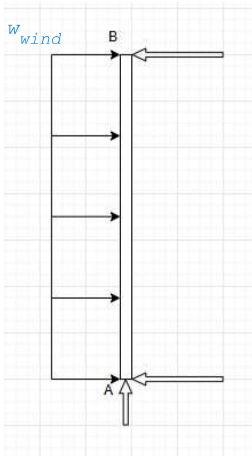
$\rho_{aliminium} := 2,7\ \frac{\mathrm{g}}{\mathrm{m}^3}$ Density of aluminium from Macsteel



$P := 0,5 \cdot \rho \cdot v_{wind}^2 \cdot C_d = 88,8889\ \mathrm{Pa}$ Pressure on panels due to wind loading (5.2)

$F := P \cdot A_{side} = 68,7556\ \mathrm{N}$ Total force on side panel

$w_{wind} := \frac{F}{2 \cdot h_1} = 41,9241\ \frac{\mathrm{N}}{\mathrm{m}}$ Force per meter of height due to wind



For aluminium section A-B asume that this member resists half of the force distributed across the entire side panel as a uniform load

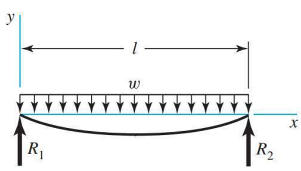
Considering the vertical bar to be simply supported and resist uniform load. The max moment occurs at the midpoint of the bar

$M_{wind} := w_{wind} \cdot \frac{h_1^2}{8} = 3,5237\ \mathrm{N\ m}$ Simple supports and uniform load from Shigleys Mechanical Design 10th edition (6.1)

$\sigma_{wind} := M_{wind} \cdot \frac{c}{I} = 7,162\ \mathrm{MPa}$ Max stress due to uniform wind load (6.2)

$y_{max} := \frac{-5 \cdot w_{wind} \cdot h_1^4}{384 \cdot E \cdot I} = -0,0007\ \mathrm{m}$ Maximum deflection at midpoint of bar (6.3)

7 Simple supports—uniform load



$$m_{frame} := 0,3\ \frac{kg}{m} \cdot \left(2 \cdot \left(h_1 + h_2 + b + \sqrt{b^2 + \left(h_1 - h_2\right)^2}\right) + 5 \cdot l\right) = 3,6233\ kg$$ Mass of all aluminium sections

$$A := l \cdot h_1 \cdot 2 + l \cdot h_2 + \left(2 \cdot b \cdot \frac{\left(h_2 - h_1\right)}{2} + 2 \cdot b \cdot h_1\right)$$ Area of all sides of perspex

$$V := A \cdot t = 0,0126\ m^3$$ Volume of all perspex

$$m_{panels} := V \cdot \rho_{perspex} = 14,9476\ kg$$

$$m_{total} := m_{frame} + m_{panels} = 18,5709\ kg$$ Total mass of structure (6.3)

$$F_{weight} := \frac{m_{total}}{4} \cdot 9,81\ \frac{m}{s^2} = 45,5451\ N$$ Asuming each vertical aluminium member supports a fourth of the total weight. (6.4)

$$\sigma_{weight} := \frac{F_{weight}}{A_{section}} = 0,5693\ MPa$$ Stress due to weight of structure (6.5)

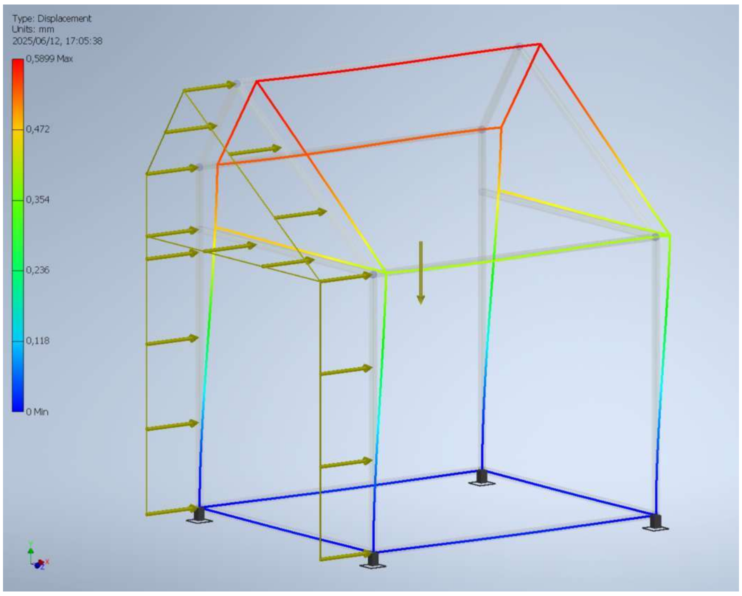
$$\sigma_{total} := \sigma_{wind} + \sigma_{weight} = 7,7314\ MPa$$ Yield strength of aluminium: 100–250 MPa → well within limits

$$S_{aliminium} := \frac{\sigma_y}{\sigma_{total}} = 40,0965$$ Factor of safety for horizontal aluminium sections under wind loading

6.1 Frame Analysis Using Inventor Enviroment:

$$Load_{wind} := 0,0038\ N\ m$$ For continuous load of 18N over all 4739mm of tubing on the side profile due to a 20km/h wind acting perpindicular to the panel. asuming 4 fixed points at the base of the greenhouse. asuming the force is resisted equally by each member on the side of the greenhouse for use in frame analysis simulation.

Maximum displacment is located at the apex of the greenhouse is limited to only 0.51mm



7.1 Fan Sizing:

$$V := b \cdot h_2 \cdot l - \frac{(h_2 - h_1)}{2} \cdot b \cdot l = 0{,}7735\ \mathrm{m}^3$$ Total volume of greenhouse

$$flow := \frac{V}{\mathrm{min}} = 0{,}7735\ \frac{\mathrm{m}^3}{\mathrm{min}}$$ Total flow required by fans using standard airflow sizing of one entire volume exchange per minute (7.1)

$$CFM := flow \cdot 35\ \frac{\mathrm{ft}^3}{\mathrm{m}^3} = 27{,}0725\ \frac{\mathrm{ft}^3}{\mathrm{min}}$$ Total flow in CFM

7.2 HAF Requirements:

$$A_{floor} := b \cdot l = 0{,}85\ \mathrm{m}^2$$ Total usable area of greenhouse

$$A_{floorft} := A_{floor} \cdot 10{,}764\ \frac{\mathrm{ft}^2}{\mathrm{m}^2} = 9{,}1494\ \mathrm{ft}^2$$

$$CFM_{HAF} := A_{floorft} \cdot 2\ \frac{\mathrm{ft}}{\mathrm{min}} = 18{,}2988\ \frac{\mathrm{ft}^3}{\mathrm{min}}$$ CFM Required using standard HAF sizing of 2CFM fan capacity per square foot of greenhpouse area (7.2)

8. Power Supply:

8.1 Battery sizing:

Components	Power	Usage / 24 hr	Wh / 24 hr	
Heater	120 W	2 hr	240 Whr	~2 hours (nighttime)
Grow − Light	100 W	3 hr	300 Whr	~3 hours (morning/afternoon)
Ventilation − Fans	10 W	8 hr	80 Whr	~8 hours (daytime)
Sensors − Actuators − Controller	10 W	24 hr	240 Whr	~24 hours (intermittent use)
TOTAL	▮	▮	860 Whr	

$Draw := 860\ \mathrm{W\ hr}$

$Voltage := 12\ \mathrm{V}$

$DoD := 80\ \%$ Allowable Depth of Discharge

$$Capacity := \frac{Draw}{Voltage \cdot DoD} = 89{,}5833\ \mathrm{A\ hr}$$ Minimum required battery capacity (8.1)

8.2 Solar Panel Sizing:

$Battery_{Ah} := 100\ \mathrm{A\ hr}$ Selecting a 100Ah battery

$hours_{day} := 6\ \mathrm{hr}$ Asuming 6 hours of sunlight per day

$$W_{panel} := Battery_{Ah} \cdot \frac{Voltage}{hours_{day} \cdot 0{,}8} = 250\ \mathrm{W}$$ Asuming effeciency of 80% (8.2)

A solar array rated to provide atleast 250W would be ableto charge the battery fully in one day asuming a moderate system effeciency of 80%

Appendix H: Greenhouse Setup Images

(a) MPPT Controller

(b) Control Room

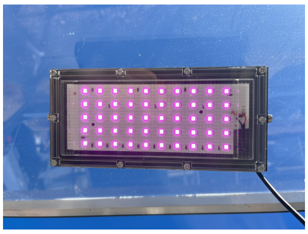
(c) Soil Sensor

(d) PV Panels

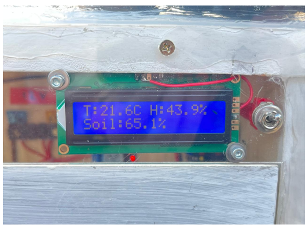
(e) Side with Active Vent

(f) Setup Overview

(g) Active Misting System



(h) LED Lighting



(i) Local Monitoring

Figure H.1: Overview of greenhouse hardware components and setup.

Appendix I: Responsible use of resources and an end-of-life strategy

The responsible use of resources and a well-defined end-of-life strategy are crucial to the project. Without a sustainable plan for decommissioning, the benefits of the project, such as advancing research in renewable energy and promoting food sustainability through small-scale greenhouses, could be overshadowed by the environmental impact of improperly discarded materials. During the design and construction phase of the project, opportunities to reuse previously used materials or use recycled material were taken into consideration.

I.1 Resource Consumption Overview

The greenhouse consisted of several integrated subsystems. The physical structure, made from an aluminium frame and Perspex panels, provided an enclosed environment necessary to establish a stable greenhouse microclimate.

The system incorporated multiple electronic components, including sensors, a microcontroller, actuators, and supporting wiring and interface elements, to enable precise control of internal conditions. Water was used for irrigation, providing accurate regulation of soil moisture, humidity, and temperature through controlled misting.

Electrical power was supplied by a solar panel, battery, and charge controller, ensuring fully off-grid operation powered entirely by renewable energy sources.

I.2 Strategies for Resource Conservation and Environmental Impact Reduction

The greenhouse was designed to operate entirely on solar energy, relying solely on electricity generated from renewable resources. Component selection prioritized energy efficiency to ensure low resource consumption and long-term compatibility with off-grid solar operation.

Sensor feedback allowed actuators to operate only when necessary, conserving power and extending actuator lifespan. Although mains water was used to supply the misting system with sufficient pressure, smart control logic implemented a duty-cycle misting pattern. This ensured short, precise watering phases, preventing overwatering caused by delayed sensor feedback or control timing. The misting nozzles were positioned to target specific areas of the growth medium, further improving water efficiency.

These strategies ensured that only the required amount of water was used, a crucial consideration in a water-scarce climate like South Africa. The ventilation system also emphasized passive ventilation, using the natural buoyancy effect to exhaust warm air, see appendix G. fans were activated only when necessary to assist in airflow requirements in conjuction with the passive ventilation techniques.

Finally, durable and recyclable materials were selected to extend component life, reduce maintenance and replacement requirements, and support material reuse in future projects. These design choices align with principles of responsible resource use and sustainable end-of-life management.

I.3 End-of-Life Strategies

Once completed, the greenhouse can serve as a model to demonstrate the potential of engineering projects, inspire the adoption of renewable energy solutions, and promote education on food sustainability and self-sufficiency. Once this post completion life cycle is over. The Project can be broken down and stripped of its parts. Some of the parts will go into storage to be reused for future engineering Thesis topics or other experiments to be used by the university at their own discretion. Other parts that are less likely to be reused are to be thrown out responsibly or recycled where possible. Special attention should be paid to proper disposal of hazardous materials.